Chapter 10
Inspection Concepts & Techniques

Inspections are visual examinations and manual checks to determine the condition of an aircraft or component. An aircraft inspection can range from a casual walk around to a detailed inspection involving complete disassembly and the use of complex inspection aids.

An inspection system consists of several processes, including reports made by mechanics, the pilot, or crew flying an aircraft and regularly scheduled inspections of an aircraft. An inspection system is designed to maintain an aircraft in the best possible condition. Thorough and repeated inspections must be considered the backbone of a good maintenance program. Irregular and haphazard inspections invariably result in gradual and certain deterioration of an aircraft. The time spent repairing an abused aircraft often totals far more than any time saved in hurrying through routine inspections and maintenance.

It has been proven that regularly scheduled inspections and preventive maintenance assure airworthiness. Operating failures and malfunctions of equipment are appreciably reduced if excessive wear or minor defects are detected and corrected early. The importance of inspections and the proper use of records concerning these inspections cannot be overemphasized.

Airframe and engine inspections may range from preflight inspections to detailed inspections. The time intervals for the inspection periods vary with the models of aircraft involved and the types of operations being conducted. The airframe and engine manufacturer's instructions should be consulted when establishing inspection intervals.

Aircraft may be inspected using a flight hours inspection system, a calendar inspection system, or a combination of both. Under the calendar inspection system, the appropriate inspection is performed on the expiration of a specified number of calendar weeks. The calendar inspection system is an efficient system from a maintenance management standpoint. Scheduled replacement of components with stated hourly operating limitations is normally accomplished during the calendar inspection falling nearest the hourly limitation. In some instances, a flight hour limitation is established to limit the number of hours that may be flown during the calendar interval.

Aircraft operating under the flight hour system are inspected when a specified number of flight hours are accumulated. Components with stated hourly operating limitations are normally replaced during the inspection that falls nearest the hourly limitation.

Basic Inspection

Techniques/Practices

Before starting an inspection, be certain all plates, access doors, fairings, and cowling have been opened or removed and the structure cleaned. When opening inspection plates and cowling, and before cleaning the area, take note of any oil or other evidence of fluid leakage.

Preparation

In order to conduct a thorough inspection, a great deal of paperwork and/or reference information must be accessed and studied before proceeding to the aircraft to conduct the inspection. The aircraft logbooks must be reviewed to provide background information and a maintenance history of the particular aircraft. The appropriate checklist or checklists must be utilized to ensure that no items are forgotten or overlooked during the inspection. Also, many additional publications must be available, either in hard copy or in electronic format, to assist in the inspections. These additional publications may include information provided by the aircraft and engine manufacturers, appliance manufacturers, parts vendors, and the Federal Aviation Administration (FAA).

Aircraft Logs

"Aircraft logs," as used in this handbook, is an inclusive term that applies to the aircraft logbook and all supplemental records concerned with the aircraft. They may come in a variety of formats. For a small aircraft, the log may indeed be a small 5" × 8" logbook. For larger aircraft, the logbooks are often larger and in the form of a three-ring binder. Aircraft that have been in service for a long time are likely to have several logbooks.

The aircraft logbook is the record where all data concerning the aircraft is recorded. Information gathered in this log is used to determine the aircraft condition, date of inspections, time on airframe, engines, and propellers. It reflects a

history of all significant events occurring to the aircraft, its components, and accessories. Additionally, it provides a place for indicating compliance with FAA airworthiness directives (ADs) or manufacturers' service bulletins (SB). The more comprehensive the logbook, the easier it is to understand the aircraft's maintenance history.

When the inspections are completed, appropriate entries must be made in the aircraft logbook certifying that the aircraft is in an airworthy condition and may be returned to service. When making logbook entries, exercise special care to ensure that the entry can be clearly understood by anyone having a need to read it in the future. Also, if making a hand-written entry, use good penmanship and write legibly. To some degree, the organization, comprehensiveness, and appearance of the aircraft logbooks have an impact on the value of the aircraft. High quality logbooks can mean a higher value for the aircraft.

Checklists

Always use a checklist when performing an inspection. The checklist may be of your own design, one provided by the manufacturer of the equipment being inspected, or one obtained from some other source. The checklist should include the following:

1. Fuselage and Hull Group
    a. Fabric and skin—for deterioration, distortion, other evidence of failure, and defective or insecure attachment of fittings.
    b. Systems and components—for proper installation, apparent defects, and satisfactory operation.
    c. Envelope gas bags, ballast tanks, and related parts—for condition.
2. Cabin and Cockpit Group
    a. General—for cleanliness and loose equipment that needs to be secured.
    b. Seats and safety belts—for condition and security.
    c. Windows and windshields—for deterioration and breakage.
    d. Instruments—for condition, mounting, marking, and (where practicable) for proper operation.
    e. Flight and engine controls—for proper installation and operation.
    f. Batteries—for proper installation and charge.
    g. All systems—for proper installation, general condition, apparent defects, and security of attachment.
3. Engine and Nacelle Group
    a. Engine section—for visual evidence of excessive oil, fuel, hydraulic leaks, and sources of such leaks.
    b. Studs and nuts—for proper torquing and obvious defects.
    c. Internal engine—for cylinder compression and for metal particles or foreign matter on screens and sump drain plugs. If cylinder compression is weak, check for improper internal condition and improper internal tolerances.
    d. Engine mount—for cracks and looseness of mounting.
    e. Flexible vibration dampeners—for condition and deterioration.
    f. Engine controls—for defects, proper travel, and proper safetying.
    g. Lines, hoses, and clamps—for leaks, condition, and looseness.
    h. Exhaust stacks—for cracks, defects, and proper attachment.
    i. Accessories—for apparent defects in security of mounting.
    j. All systems—for proper installation, general condition defects, and secure attachment.
    k. Cowling—for cracks and defects.
    l. Ground run-up and functional check—check all powerplant controls and systems for correct response, all instruments for proper operation and indication.
4. Landing Gear Group
    a. All units—for condition and security of attachment.
    b. Shock absorbing devices—for proper oleo fluid level.
    c. Linkage, trusses, and members—for undue or excessive wear, fatigue, and distortion.
    d. Retracting and locking mechanism—for proper operation.
    e. Hydraulic lines—for leakage.
    f. Electrical system—for chafing and proper operation of switches.
    g. Wheels—for cracks, defects, and condition of bearings.
    h. Tires—for wear and cuts.
    i. Brakes—for proper adjustment.
    j. Floats and skis—for security of attachment and

obvious defects.

5. Wing and Center Section
   a. All components—for condition and security.
   b. Fabric and skin—for deterioration, distortion, other evidence of failure, and security of attachment.
   c. Internal structure (spars, ribs, compression members)—for cracks, bends, and security.
   d. Movable surfaces—for damage or obvious defects, unsatisfactory fabric or skin attachment, and proper travel.
   e. Control mechanism—for freedom of movement, alignment, and security.
   f. Control cables—for proper tension, fraying, wear, and proper routing through fairleads and pulleys.
6. Empennage Group
   a. Fixed surfaces—for damage or obvious defects, loose fasteners, and security of attachment.
   b. Movable control surfaces—for damage or obvious defects, loose fasteners, loose fabric, or skin distortion.
   c. Fabric or skin—for abrasion, tears, cuts, defects, distortion, and deterioration.
7. Propeller Group
   a. Propeller assembly—for cracks, nicks, bends, and oil leakage.
   b. Bolts—for proper torquing and safe tying.
   c. Anti-icing devices—for proper operation and obvious defects.
   d. Control mechanisms—for proper operation, secure mounting, and travel.
8. Communication and Navigation Group
   a. Radio and electronic equipment—for proper installation and secure mounting.
   b. Wiring and conduits—for proper routing, secure mounting, and obvious defects.
   c. Bonding and shielding—for proper installation and condition.
   d. Antennas—for condition, secure mounting, and proper operation.
9. Miscellaneous
   a. Emergency and first aid equipment—for general condition and proper stowage.
   b. Parachutes, life rafts, flares, and so forth—inspect in accordance with the manufacturer's recommendations.
   c. Autopilot system—for general condition, security of attachment, and proper operation.

Publications

Aeronautical publications are the sources of information for guiding aviation mechanics in the operation and maintenance of aircraft and related equipment. The proper use of these publications greatly aid in the efficient operation and maintenance of all aircraft. These include manufacturers' SBs, manuals, and catalogs; FAA regulations; ADs; advisory circulars (ACs); and aircraft, engine, and propeller specifications.

Manufacturers' Service Bulletins/Instructions

Service bulletins or service instructions are two of several types of publications issued by airframe, engine, and component manufacturers. The bulletins may include: purpose for issuing the publication; name of the applicable airframe, engine, or component; detailed instructions for service, adjustment, modification or inspection, and source of parts, if required; and estimated number of man-hours required to accomplish the job.

Maintenance Manual

The manufacturer's aircraft maintenance manual contains complete instructions for maintenance of all systems and components installed in the aircraft. It contains information for the mechanic who normally works on components, assemblies, and systems while they are installed in the aircraft, but not for the overhaul mechanic. A typical aircraft maintenance manual contains:

- A description of the systems (i.e., electrical, hydraulic, fuel, control)
- Lubrication instructions setting forth the frequency and the lubricants and fluids that are to be used in the various systems
- Pressures and electrical loads applicable to the various systems
- Tolerances and adjustments necessary to proper functioning of the airplane
- Methods of leveling, raising, and towing
- Methods of balancing control surfaces
- Identification of primary and secondary structures
- Frequency and extent of inspections necessary to the proper operation of the airplane

- Special repair methods applicable to the airplane
- Special inspection techniques requiring x-ray, ultrasonic, or magnetic particle inspection
- A list of special tools

Overhaul Manual

The manufacturer's overhaul manual contains brief descriptive information and detailed step-by-step instructions covering work normally performed on a unit that has been removed from the aircraft. Simple, inexpensive items, such as switches and relays where overhaul is uneconomical, are not covered in the overhaul manual.

Structural Repair Manual

The structural repair manual contains the manufacturer's information and specific instructions for repairing primary and secondary structures. Typical skin, frame, rib, and stringer repairs are covered in this manual. Also, included are material and fastener substitutions and special repair techniques.

Illustrated Parts Catalog

The illustrated parts catalog presents component breakdowns of structure and equipment in disassembly sequence. Also, included are exploded views or cutaway illustrations for all parts and equipment manufactured by the aircraft manufacturer.

Wiring Diagram Manual

The wiring diagram manual is a collection of diagrams, drawings, and lists that define the wiring and hook up of associated equipment installed on airplanes. The data is organized in accordance with the Air Transport Association ATA iSPec 2200 specification.

Code of Federal Regulations (CFRs)

The Code of Federal Regulations (CFRs) were established by law to provide for the safe and orderly conduct of flight operations and to prescribe airmen privileges and limitations. A knowledge of the CFRs is necessary during the performance of maintenance, since all work done on aircraft must comply with CFR provisions.

Airworthiness Directives (ADs)

A primary safety function of the FAA is to require correction of unsafe conditions found in an aircraft, aircraft engine, propeller, or appliance when such conditions exist and are likely to exist or develop in other products of the same design. The unsafe condition may exist because of a design defect, maintenance, or other causes. Title 14 of the CFR part 39, Airworthiness Directives, defines the authority and responsibility of the administrator for requiring the necessary corrective action. The ADs are published to notify aircraft owners and other interested persons of unsafe conditions and to prescribe the conditions that the product may continue to be operated. Furthermore, these are federal aviation regulations and must be complied with unless specific exemption is granted.

There are two categories of ADs:

1. Those of an emergency nature requiring immediate compliance upon receipt.
2. Those of a less urgent nature requiring compliance within a relatively longer period of time.

Also, ADs may be a one-time compliance item or a recurring item that requires future inspection on an hourly basis (accrued flight time since last compliance) or a calendar time basis.

The contents of ADs include the aircraft, engine, propeller, or appliance model and serial numbers affected. Also, included are the compliance time or period, a description of the difficulty experienced, and the necessary corrective action.

Type Certificate Data Sheets (TCDS)

The type certificate data sheet (TCDS) describes the type design and sets forth the limitations prescribed by the applicable CFR part. It also includes any other limitations and information found necessary for type certification of a particular model aircraft. *[Figure 10-1]*

All TCDS are numbered in the upper right corner of each page. This number is the same as the type certificate number. The name of the type certificate holder, together with all of the approved models, appears immediately below the type certificate number. The issue date completes this group. This information is contained within a bordered text box to set it off.

The TCDS is separated into one or more sections. Each section is identified by a Roman numeral followed by the model designation of the aircraft that the section pertains. The category or categories that the aircraft can be certificated in are shown in parentheses following the model number. Also, included is the approval date shown on the type certificate.

The data sheet contains information regarding:

1. Model designation of all engines that the aircraft manufacturer obtained approval for use with this model aircraft.
2. Minimum fuel grade to be used.
3. Maximum continuous and takeoff ratings of the approved engines, including manifold pressure (when used), rotations per minute (rpm), and horsepower (hp).
4. Name of the manufacturer and model designation for each propeller that the aircraft manufacturer obtained

approval is shown together with the propeller limits and any operating restrictions peculiar to the propeller or propeller engine combination.

5. Airspeed limits in both miles per hour (mph) and knots.
6. Center of gravity (CG) range for the extreme loading conditions of the aircraft is given in inches from the datum. The range may also be stated in percent of mean aerodynamic chord (%MAC) for transport category aircraft.
7. Empty weight center of gravity (EWCG) range (when established) is given as fore and aft limits in inches from the datum. If no range exists, the word "none" is shown following the heading on the data sheet.
8. Location of the datum.
9. Means provided for leveling the aircraft.
10. All pertinent maximum weights.
11. Number of seats and their moment arms.
12. Oil and fuel capacity.
13. Control surface movements.
14. Required equipment.
15. Additional or special equipment found necessary for certification.
16. Information concerning required placards.

It is not within the scope of this handbook to list all the items that can be shown on the TCDS. Those items listed above serve only to acquaint aviation mechanics with the type of information generally included on the data sheets. TCDS may be many pages in length.

When conducting a required or routine inspection, it is necessary to ensure that the aircraft and all the major items on it are as defined in the TCDS. The inspector ensures that all installed aircraft equipment conforms to the TCDS. This is called a conformity check and verifies that the aircraft conforms to the specifications of the aircraft as it was originally certified. Sometimes alterations are made that are not specified or authorized in the TCDS. When that condition exists, a supplemental type certificate (STC) is issued. STCs are considered a part of the permanent records of an aircraft and should be maintained as part of that aircraft's logs.

Routine/Required Inspections

For the purpose of determining their overall condition, 14 CFR provides for the inspection of all civil aircraft at specific intervals, depending generally upon the type of operations that they are engaged in. The pilot-in-command (PIC) of a civil aircraft is responsible for determining whether that aircraft is in a condition for safe flight. Therefore, the aircraft must be inspected before each flight. More detailed inspections must be conducted by aviation maintenance technicians (AMTs at least once each 12 calendar months, while inspection is required for others after each 100 hours of flight. In other instances, an aircraft may be inspected in accordance with a system set up to provide for total inspection of the aircraft over a calendar or flight time period. These include phase-type inspections.

To determine the specific inspection requirements and rules for the performance of inspections, refer to the CFR that prescribes the requirements for the inspection and maintenance of aircraft in various types of operations.

Preflight/Postflight Inspections

Pilots are required to follow a checklist contained within the Pilot's Operating Handbook (POH) when operating aircraft. The first section of the checklist is entitled "Preflight Inspection." The preflight inspection checklist includes a "walk-around" section listing items that the pilot is to visually check for general condition as they walk around the airplane. Also, the pilot must ensure that fuel, oil, and other items required for flight are at the proper levels and not contaminated. Additionally, it is the pilot's responsibility to review the aircraft maintenance records, and other required paperwork to verify that the aircraft is indeed airworthy. After each flight, it is recommended that the pilot or mechanic conduct a postflight inspection to detect any problems that might require repair or servicing before the next flight.

Annual/100-Hour Inspections

The basic requirements for annual and 100-hour inspections are discussed in 14 CFR part 91. With some exceptions, all aircraft must have a complete inspection annually. Aircraft that are used for commercial purposes (carrying any person, other than a crewmember, for hire or flight instruction for hire) and are likely to be used more frequently than noncommercial aircraft must have this complete inspection every 100 hours. The scope and detail of items to be included in annual and 100-hour inspections is included as Appendix D to part 43. *[Figure 10-2]*

A properly written checklist, such as the one shown earlier in this chapter, includes all the items of Appendix D. Although the scope and detail of annual and 100-hour inspections are identical, there are two significant differences. One difference involves persons authorized to conduct them. A certified airframe and powerplant (A&P) maintenance technician can conduct a 100-hour inspection, whereas an annual inspection must be conducted by a certified A&P maintenance technician with inspection authorization (IA). The other difference involves authorized overflight of the maximum 100 hours before inspection. An aircraft may be flown up to 10 hours beyond the 100-hour limit if necessary to fly to a destination

DEPARTMENT OF TRANSPORTATION
FEDERAL AVIATION ADMINISTRATION

A27EU
Revision 4
AIRBUS DEFENCE AND SPACE GMBH
EADS DEUTSCHLAND GMBH
DAIMLER CHRYSLER AEROSPACE AG
DAIMLER-BENZ AEROSPACE AG
DEUTSCHE AEROSPACE AG
MESSERSCHMITT-BÖLKOW-BLOHM AG
MESSERSCHMITT-BÖLKOW-BLOHM GMBH
BO-209-150 FV & RV
BO-209-160 FV & RV
BO-209-150 FF
July 9, 2015

TYPE CERTIFICATE DATA SHEET NO. A27EU

This data sheet, which is a part of Type Certificate No. A27EU, prescribes conditions and limitations under which the product for which the Type Certificate was issued meets the airworthiness requirements of the Federal Aviation Regulations.

Type Certificate Holder	Airbus Defence and Space GmbH Willy-Messerschmitt-Strasse 1 85521 Ottobrunn Germany
Type Certificate Ownership Record	Messerschmitt-Bölkow-Blohm GmbH transferred TC A27EU to Messerschmitt-Bölkow-Blohm AG on April 1, 1992 (See NOTE 4.) Messerschmitt-Bölkow-Blohm AG transferred TC A27EU to Deutsche Aerospace AG on November 30, 1992 Deutsche Aerospace AG transferred TC A27EU to Daimler-Benz Aerospace AG on January 2, 1995 Daimler-Benz Aerospace AG transferred TC A27EU to Daimler Chrysler Aerospace AG on November 17, 1998 Daimler Chrysler Aerospace AG transferred TC A27EU to EADS Deutschland GmbH on July 10, 2000 EADS Deutschland GmbH transferred TC A27EU to Airbus Defence and Space GmbH on July 1, 2014 (See NOTE 7.)

I - Model BO-209-150 FV and RV, 2 PCLM (Normal and Utility Category), approved 9 July 1971
(FV model has fixed nose L.g.; RV model has retractable nose L.g.).

Engine	Lycoming O-320-E1C or O-320-E1F
Fuel	80/87 minimum grade aviation gasoline
Engine limits	For all operations, 2700 r.p.m. (150 hp.)
Propeller and propeller limits	Hartzell HC-C2YL-1B/7663A-6 Diameter: 70 in. no further reduction permitted Pitch setting at 30 in. radius: High 27° Low 12°12' Spinner: MBB P/N 209-61056 Governor: Woodward P/N T210452 or P/N 210681

Page No.	1	2	3	4	5	6
Rev. No.	4	-	-	4	2	4

Figure 10-1. *Type certificate data sheet (TCDS).*

Airspeed limits (CAS)	Normal and Utility Category	
	Never exceed	173 knots (199 m.p.h.)
	Maximum structural cruising	135 knots (155 m.p.h.)
	Maneuvering	117 knots (135 m.p.h.)
	Flaps extended	88 knots (101 m.p.h.)
	*Landing gear operation	104 knots (120 m.p.h.)
	*Landing gear extended	173 knots (199 m.p.h.)
	(*Applies only to the RV model).	
C.G. range	Normal Category	
	(85.47) to (89.37) at 1265 lb. or less	
	(86.92) to (89.37) at 1808 lb.	
	Utility Category	
	(85.47) to (89.37) at 1265 lb. or less	
	(86.25) to (89.37) at 1565	
Maximum weight	1808 lb., for Normal Category	
	1565 lb., for Utility Category	
No. of seats	2 at (+ 90.7)	
Maximum baggage	110 lb. at (+114.2)	
Fuel capacity	39.2 gal. total (38.6 gal. usable; two 19.6 gal., wing tanks at + 90.7)	
Oil capacity	8 qt. (+3.94)	
	See NOTE 1 for unusable fuel and undrainable oil data.	

II - Model BO-209-160 FV and RV, 2 PCLM (Normal and Utility Category), approved 9 July 1971
(FV model has fixed nose L.g.; RV model has retractable nose L.g.).

Engine	Lycoming IO-320-D1A or IO-320-D1B	
Fuel	100/130 minimum grade aviation gasoline	
Engine limits	For all operations, 2700 r.p.m. (160 hp.)	
Propeller and propeller limits	Hartzell HC-C2YL-1B/7663A-6	
	Diameter: 70 in. no further reduction permitted	
	Pitch setting at 30 in. radius:	High 27°
		Low 14°57'
	Spinner:	MBB P/N 209-61056
	Governor:	Woodward P/N T210452 or P/N 210681
Airspeed limits (CAS)	Normal and Utility Category	
	Never exceed	173 knots (199 m.p.h.)
	Maximum structural cruising	135 knots (155 m.p.h.)
	Maneuvering	117 knots (135 m.p.h.)
	Flaps extended	88 knots (101 m.p.h.)
	*Landing gear operation	104 knots (120 m.p.h.)
	*Landing gear extended	173 knots (199 m.p.h.)
	(*Applies only to the RV model).	
C.G. range	Normal Category	
	(85.47) to (89.37) at 1265 lb. or less	
	(86.92) to (89.37) at 1808 lb.	
	Utility Category	
	(85.47) to (89.37) at 1265 lb. or less	
	(86.25) to (89.37) at 1565 lb.	

Figure 10-1. *Type certificate data sheet (TCDS) (continued).*

Maximum weight	1808 lb. for Normal Category 1565 lb. for Utility Category
No. of seats	2 at (+ 90.7)
Maximum baggage	110 lb. at (+ 114.2)
Fuel capacity	39.2 gal. total (38.6 gal. usable; two 19.6 gal., wing tanks at + 90.7)
Oil capacity	8 qt. (+3.94) See NOTE 1 for unusable fuel and undrainable oil data.

III - Model BO-209-150 FF, 2 PCLM (Normal and Utility Category), approved 9 July 1971 (fixed nose L.g.).

Engine Lycoming O-320-E2C or O-320-E2F

Fuel 80/87 minimum grade aviation gasoline.

Engine limits	For all operations, 2700 r.p.m. (150 hp.)
Propeller and propeller limits	McCauley 1C172MGM-70.5-60 or -66 Static r.p.m. at maximum permissible throttle setting: Not over 2400, not under 2100 No additional tolerance permitted. Diameter: Maximum 70.5 in., minimum for repairs 70 in. No further reduction permitted Spinner: MBB P/N 209-61156

Airspeed limits (CAS)

Normal and Utility Category

Never exceed	173 knots (199 m.p.h.)
Maximum structural cruising	135 knots (155 m.p.h.)
Maneuvering	117 knots (135 m.p.h.)
Flaps extended	88 knots (101 m.p.h.)

C.G. range	Normal Category (85.47) to (89.37) at 1265 lb. or less (86.92) to (89.37) at 1808 lb. Utility Category (85.47) to (89.37) at 1265 lb. or less (86.25) to (89.37) at 1565 lb.
Maximum weight	1808 lb., for Normal Category 1565 lb., for Utility Category
No. of seats	2 at (+ 90.7)
Maximum baggage	110 lb. at (+114.2)
Fuel capacity	39.2 gal. total (38.6 gal. usable; two 19.6 gal., wing tanks at + 90.7)
Oil capacity	8 qt. (+3.94) See NOTE 1 for unusable fuel and undrainable oil data.

Figure 10-1. *Type certificate data sheet (TCDS) (continued).*

DATA PERTINENT TO ALL MODELS

Control Surface Movements	Ailerons	Up	29° ± 1°	Down	14° ± 1°
	Wing flaps			Down	35° + 0° - 3°
	Stabilator	Up	18° ± 1°	Down	9° ± 1°
	Rudder	Left	28° ± 2°	Right	28° ± 2°

Stabilator trim, distance measured between trailing edge of trim tab and trailing edge of stabilator with stabilator in the neutral position.

tab neutral:	0.32 in. Down,	± 0.08 in.
nose down:	0.20 in. Up,	± 0.08 in.
nose up:	0.66 in. Down,	± 0.08 in.
total travel:	0.86 in.	± 0.16 in.

Datum

75.51 in. forward of wing leading edge at split line of the wing/wing stub fairing.

Leveling means

Two leveling points on left side of fuselage.

Serial Nos. eligible

Serial Numbers 121 and subsequent.
The Federal Republic of Germany Government Certificate of Airworthiness for Export endorsed as noted below under "Import Requirements" must be submitted for each individual aircraft for which application for airworthiness certification is made.

Certification basis

FAR 21.29 and FAR 23 dated 1 February 1965 as amended by Amendments 23-1 through 23-9 inclusive. Type Certificate No. A27EU, issued 9 July 1971.
Date of Application for Type Certificate: 11 May 1970.

The Luftfahrt Bundesamt originally type certificated this aircraft under its type certificate Number 680. The FAA validated this product under U.S. Type Certificate Number A27EU. Effective September 28, 2003, the European Aviation Safety Agency (EASA) began oversight of this product on behalf of Germany.

The EASA type certificate for the BO-209 models is EASA.A.357.

Import Requirements

The FAA can issue a U.S. airworthiness certificate based on an NAA Export Certificate of Airworthiness (Export C of A) signed by a representative of the Luftfahrt Bundesamt on behalf of the European Community. The Export C of A should contain the following statement: 'The aircraft covered by this certificate has been examined, tested, and found to comply with U.S. airworthiness regulations 14 CFR Part 23 approved under U.S. Type Certificate No. A27EU and to be in a condition for safe operation.'

Service Information

Each of the documents listed below must state that it is approved by the European Aviation Safety Agency (EASA) or – for approvals made before September 28, 2003 – by the Luftfahrt Bundesamt.

- Service bulletins,
- Structural repair manuals,
- Vendor manuals,
- Aircraft flight manuals, and
- Overhaul and maintenance manuals.

The FAA accepts such documents and considers them FAA-approved unless one of the following conditions exists:

- The documents change the limitations, performance, or procedures of the FAA approved manuals; or

- The documents make an acoustical or emissions changes to this product's U.S. type certificate as defined in 14 CFR § 21.93.

Figure 10-1. *Type certificate data sheet (TCDS) (continued).*

Service Information, cont'd	The FAA uses the post type validation procedures to approve these documents. The FAA may delegate on case-by-case to EASA to approve on behalf of the FAA for the U.S. type certificate. If this is the case it will be noted on the document.
Equipment	The basic required equipment as prescribed in the applicable airworthiness regulations (see Certification Basis) must be installed in the aircraft for certification. In addition, the following items of equipment are required: 1. Stall Warning System. 2. LBA-approved Model BO-209 Approved Flight Manual, Ref. No.LF 37E-7/71 dated July 1971 or later LBA-approved revision. 3. Airplanes S/N 121 through 130 must be modified in accordance with MBB Technical Note TN 9-71 to provide an alternate static system source and an aural landing gear warning system. (These systems are incorporated in production on S/Ns 131 and subsequent).

NOTE 1. Current weight and balance report including list of equipment in certificated empty weight, and loading instructions when necessary, must be provided for each airplane at the time of original airworthiness certification. The certificated empty weight and corresponding center of gravity must include undrainable oil of 0 lbs. at +39.4 and unusable fuel of 3.6 lb. at +90.7.

NOTE 2. The following placard must be displayed in front and in clear view of the pilot:

"This airplane must be operated as a Normal or Utility Category airplane in compliance with their operating limitations stated in the form of placards, markings, and manuals."

In addition, all placards required in the LBA-approved Airplane Flight Manual must be installed in the appropriate location.

NOTE 3. Information essential for proper maintenance of the airplane is contained in the Messerschmitt-Bolkow-Blohm GmbH., Model BO-209 Maintenance Manual included in MBB document Ref. LF 37E-7/71.

NOTE 4. The airplane manufacturer is:

Waggon- und Maschienenbau A.G.
Donauworth, Laupheim
Federal Republic of Germany
(A division of Messerschmitt-Bolkow-Blohm).

NOTE 5. Installation of a Tost tow coupling (ring type), LBA approval No. 60.230.4 may be approved when installed in accordance with MBB Drwg. 209-85003 (for glider towing) or MBB Drwgs. 209-85003 and 209-8700 (for banner towing).

NOTE 6. For issuance of an airworthiness certificate in accordance with 14 CFR Part 21.182(c), the Luftfahrt Bundesamt of Germany must certify that the airplane conforms to the type design and is in a condition for safe operation. In that regard, the Luftfahrt Bundesamt of Germany will certify that the airplane complies with all applicable mandatory continuing airworthiness information (MCAI) it has issued. For issuance of an airworthiness certificate in accordance with 14 CFR Part 21.182(d) the certificating inspector, or other authorized person, must find, among other things, that the product is in a condition for safe operation. In order to make that finding, the certificating inspector or other authorized person should contact ACE-112, Federal Aviation Administration, Small Airplane Directorate, prior to issuance to determine whether showing airplane compliance with certain MCAI is necessary to support a finding that the airplane is in a condition for safe operation.

Figure 10-1. *Type certificate data sheet (TCDS) (continued).*

NOTE 7. Some of these transfers were not notified to the FAA and so in some instances the actual type certificates were not reissued.

.....END.....

Figure 10-1. *Type certificate data sheet (TCDS) (continued).*

where the inspection is to be conducted.

Progressive Inspections

Because the scope and detail of an annual inspection is very extensive and could keep an aircraft out of service for a considerable length of time, alternative inspection programs designed to minimize down time may be utilized. A progressive inspection program allows an aircraft to be inspected progressively. The scope and detail of an annual inspection is essentially divided into segments or phases (typically four to six). Completion of all the phases completes a cycle that satisfies the requirements of an annual inspection. The advantage of such a program is that any required segment may be completed overnight and thus enable the aircraft to fly daily without missing any revenue earning potential. Progressive inspection programs include routine items, such as engine oil changes, and detailed items, such as flight control cable inspection. Routine items are accomplished each time the aircraft comes in for a phase inspection, and detailed items focus on detailed inspection of specific areas. Detailed inspections are typically done once each cycle. A cycle must be completed within 12 months. If all required phases are not completed within 12 months, the remaining phase inspections must be conducted before the end of the 12th month from when the first phase was completed.

Each registered owner or operator of an aircraft desiring to use a progressive inspection program must submit a written request to the FAA Flight Standards District Office (FSDO) having jurisdiction over the area that the applicant is located. Section 91.409(d) of 14 CFR part 91 establishes procedures to be followed for progressive inspections. *[Figure 10-3]*

Continuous Inspections

Continuous inspection programs are similar to progressive inspection programs, except that they apply to large or turbine-powered aircraft and are therefore more complicated. Like progressive inspection programs, they require approval by the FAA Administrator. The approval may be sought based upon the type of operation and the CFR parts that the aircraft is operated under. The maintenance program for commercially operated aircraft must be detailed in the approved operations specifications (OpSpecs) of the commercial certificate holder.

Airlines utilize a continuous maintenance program that includes both routine and detailed inspections. However, the detailed inspections may include different levels of detail. Often referred to as "checks," the A-checks, B-checks, C-checks, and D-checks involve increasing levels of detail. A-checks are the least comprehensive and occur frequently. D-checks, on the other hand, are extremely comprehensive, involving major disassembly, removal, overhaul, and inspection of systems and components. They might occur only three to six times during the service life of an aircraft.

Altimeter & Transponder Inspections

Aircraft that are operated in controlled airspace under instrument flight rules (IFR) must have each altimeter and static system tested in accordance with procedures described in 14 CFR part 43, Appendix E, within the preceding 24 calendar months. Aircraft having an air traffic control (ATC) transponder must also have each transponder checked within the preceding 24 months. All these checks must be conducted by appropriately certified individuals.

Air Transport Association iSpec 2200

In an effort to standardize the format in which maintenance information is presented in aircraft maintenance manuals, Air Transport Association (now Airlines for America)issued specifications for Manufacturers' Technical Data. The original specification was called ATA Spec 100. Over the years, Spec 100 has been continuously revised and updated. Eventually, ATA Spec 2100 was developed for electronic documentation. These two specifications evolved into one document called ATA iSpec 2200, developed and managed by the ATA e-Business Program, a consensus-based industry standards organization administered by Airlines for America (A4A). As a result of this standardization, maintenance technicians can always find information regarding a particular system in the same section of an aircraft maintenance manual, regardless of manufacturer. For example, if seeking information about the electrical system on any aircraft, that information is always found in section (chapter) 24.

The ATA iSpec 2200 divides the aircraft into systems, such as air conditioning, that covers the basic air conditioning system (ATA 21). Numbering in each major system provides an arrangement for breaking the system down into several subsystems. *[Figure 10-4]* Late model aircraft, both over and under the 12,500-pound designation, have their parts manuals and maintenance manuals arranged according to the ATA-coded system. The following abbreviated table of ATA System, Subsystem, and Titles is included for familiarization purposes.

Keep in mind that not all aircraft have all these systems installed. Small and simple aircraft have fewer systems than larger, more complex aircraft.

Special Inspections

During the service life of an aircraft, occasions may arise when something out of the ordinary care and use of an aircraft could possibly affect its airworthiness. When these situations are encountered, special inspection procedures, also called conditional inspections, are followed to determine if damage to the aircraft structure has occurred. The procedures

Appendix D to Part 43—Scope and Detail of Items (as Applicable to the Particular Aircraft) To Be Included in Annual and 100-Hour Inspections

(a) Each person performing an annual or 100-hour inspection shall, before that inspection, remove or open all necessary inspection plates, access doors, fairing, and cowling. He shall thoroughly clean the aircraft and aircraft engine.

(b) Each person performing an annual or 100-hour inspection shall inspect (where applicable) the following components of the fuselage and hull group:

(1) Fabric and skin—for deterioration, distortion, other evidence of failure, and defective or insecure attachment of fittings.

(2) Systems and components—for improper installation, apparent defects, and unsatisfactory operation.

(3) Envelope, gas bags, ballast tanks, and related parts—for poor condition.

(c) Each person performing an annual or 100-hour inspection shall inspect (where applicable) the following components of the cabin and cockpit group:

(1) Generally—for uncleanliness and loose equipment that might foul the controls.

(2) Seats and safety belts—for poor condition and apparent defects.

(3) Windows and windshields—for deterioration and breakage.

(4) Instruments—for poor condition, mounting, marking, and (where practicable) improper operation.

(5) Flight and engine controls—for improper installation and improper operation.

(6) Batteries—for improper installation and improper charge.

(7) All systems—for improper installation, poor general condition, apparent and obvious defects, and insecurity of attachment.

(d) Each person performing an annual or 100-hour inspection shall inspect (where applicable) components of the engine and nacelle group as follows:

(1) Engine section—for visual evidence of excessive oil, fuel, or hydraulic leaks, and sources of such leaks.

(2) Studs and nuts—for improper torquing and obvious defects.

(3) Internal engine—for cylinder compression and for metal particles or foreign matter on screens and sump drain plugs. If there is weak cylinder compression, for improper internal condition and improper internal tolerances.

(4) Engine mount—for cracks, looseness of mounting, and looseness of engine to mount.

(5) Flexible vibration dampeners—for poor condition and deterioration.

(6) Engine controls—for defects, improper travel, and improper safetying.

(7) Lines, hoses, and clamps—for leaks, improper condition and looseness.

(8) Exhaust stacks—for cracks, defects, and improper attachment.

(9) Accessories—for apparent defects in security of mounting.

(10) All systems—for improper installation, poor general condition, defects, and insecure attachment.

(11) Cowling—for cracks, and defects.

(e) Each person performing an annual or 100-hour inspection shall inspect (where applicable) the following components of the landing gear group:

(1) All units—for poor condition and insecurity of attachment.

(2) Shock absorbing devices—for improper oleo fluid level.

(3) Linkages, trusses, and members—for undue or excessive wear fatigue, and distortion.

(4) Retracting and locking mechanism—for improper operation.

(5) Hydraulic lines—for leakage.

(6) Electrical system—for chafing and improper operation of switches.

(7) Wheels—for cracks, defects, and condition of bearings.

(8) Tires—for wear and cuts.

(9) Brakes—for improper adjustment.

(10) Floats and skis—for insecure attachment and obvious or apparent defects.

(f) Each person performing an annual or 100-hour inspection shall inspect (where applicable) all components of the wing and center section assembly for poor general condition, fabric or skin deterioration, distortion, evidence of failure, and insecurity of attachment.

(g) Each person performing an annual or 100-hour inspection shall inspect (where applicable) all components and systems that make up the complete empennage assembly for poor general condition, fabric or skin deterioration, distortion, evidence of failure, insecure attachment, improper component installation, and improper component operation.

(h) Each person performing an annual or 100-hour inspection shall inspect (where applicable) the following components of the propeller group:

(1) Propeller assembly—for cracks, nicks, binds, and oil leakage.

(2) Bolts—for improper torquing and lack of safetying.

(3) Anti-icing devices—for improper operations and obvious defects.

(4) Control mechanisms—for improper operation, insecure mounting, and restricted travel.

(i) Each person performing an annual or 100-hour inspection shall inspect (where applicable) the following components of the radio group:

(1) Radio and electronic equipment—for improper installation and insecure mounting.

(2) Wiring and conduits—for improper routing, insecure mounting, and obvious defects.

(3) Bonding and shielding—for improper installation and poor condition.

(4) Antenna including trailing antenna—for poor condition, insecure mounting, and improper operation.

(j) Each person performing an annual or 100-hour inspection shall inspect (where applicable) each installed miscellaneous item that is not otherwise covered by this listing for improper installation and improper operation.

Figure 10-2. *Title 14 CFR Appendix D to Part 43—Scope and detail of items (as applicable to the particular aircraft) to be included in annual and 100-hour inspections.*

§ 91.409 Inspections.

(d) Progressive inspection. Each registered owner or operator of an aircraft desiring to use a progressive inspection program must submit a written request to the FAA Flight Standards district office having jurisdiction over the area in which the applicant is located, and shall provide—

(1) A certificated mechanic holding an inspection authorization, a certificated airframe repair station, or the manufacturer of the aircraft to supervise or conduct the progressive inspection;

(2) A current inspection procedures manual available and readily understandable to pilot and maintenance personnel containing, in detail—

(i) An explanation of the progressive inspection, including the continuity of inspection responsibility, the making of reports, and the keeping of records and technical reference material;

(ii) An inspection schedule, specifying the intervals in hours or days when routine and detailed inspections will be performed and including instructions for exceeding an inspection interval by not more than 10 hours while en route and for changing an inspection interval because of service experience;

(iii) Sample routine and detailed inspection forms and instructions for their use; and

(iv) Sample reports and records and instructions for their use;

(3) Enough housing and equipment for necessary disassembly and proper inspection of the aircraft; and

(4) Appropriate current technical information for the aircraft.

The frequency and detail of the progressive inspection shall provide for the complete inspection of the aircraft within each 12 calendar months and be consistent with the manufacturer's recommendations, field service experience, and the kind of operation in which the aircraft is engaged. The progressive inspection schedule must ensure that the aircraft, at all times, will be airworthy and will conform to all applicable FAA aircraft specifications, type certificate data sheets, airworthiness directives, and other approved data. If the progressive inspection is discontinued, the owner or operator shall immediately notify the local FAA Flight Standards district office, in writing, of the discontinuance. After the discontinuance, the first annual inspection under §91.409(a)(1) is due within 12 calendar months after the last complete inspection of the aircraft under the progressive inspection. The 100-hour inspection under §91.409(b) is due within 100 hours after that complete inspection. A complete inspection of the aircraft, for the purpose of determining when the annual and 100-hour inspections are due, requires a detailed inspection of the aircraft and all its components in accordance with the progressive inspection. A routine inspection of the aircraft and a detailed inspection of several components is not considered to be a complete inspection.

Figure 10-3. *Title 14 CFR Section 91.409(d), Progressive Inspection.*

outlined on the following pages are general in nature and are intended to acquaint the aviation mechanic with the areas to be inspected. As such, they are not all inclusive. When performing any of these special inspections, always follow the detailed procedures in the aircraft maintenance manual. In situations where the manual does not adequately address the situation, seek advice from other maintenance technicians who are highly experienced with them. The following paragraphs describe some typical types of special inspections.

Hard or Overweight Landing Inspection

The structural stress induced by a landing depends not only upon the gross weight at the time, but also upon the severity of impact. The hard landing inspection is for hard landings at or below the maximum design landing limits. An overweight landing inspection must be performed when an airplane lands at a weight above the maximum design landing weight. However, because of the difficulty in estimating vertical velocity at the time of contact, it is hard to judge whether or not a landing has been sufficiently severe to cause structural damage. For this reason, a special inspection is performed after a landing is made at a weight known to exceed the design landing weight or after a rough landing, even though the latter may have occurred when the aircraft did not exceed the design landing weight.

Wrinkled wing skin is the most easily detected sign of an excessive load having been imposed during a landing. Another indication easily detected is fuel leakage along riveted seams. Other possible locations of damage are spar webs, bulkheads, nacelle skin and attachments, firewall skin, and wing and fuselage stringers. If none of these areas show adverse effects, it is reasonable to assume that no serious damage has occurred. If damage is detected, a more extensive inspection and alignment check may be necessary.

Severe Turbulence Inspection/Over "G"

When an aircraft encounters a gust condition, the airload on the wings exceeds the normal wingload supporting the aircraft weight. The gust tends to accelerate the aircraft while its inertia acts to resist this change. If the combination of gust velocity and airspeed is too severe, the induced stress can cause structural damage.

A special inspection is performed after a flight through severe turbulence. Emphasis is placed upon inspecting the upper and lower wing surfaces for excessive buckles or wrinkles with permanent set. Where wrinkles have occurred, remove a few rivets and examine the rivet shanks to determine if the rivets

have sheared or were highly loaded in shear.

Through the inspection doors and other accessible openings, inspect all spar webs from the fuselage to the tip. Check for buckling, wrinkles, and sheared attachments. Inspect for buckling in the area around the nacelles and in the nacelle skin, particularly at the wing leading edge. Check for fuel leaks. Any sizeable fuel leak is an indication that an area may have received overloads that have broken the sealant and opened the seams.

If the landing gear was lowered during a period of severe turbulence, inspect the surrounding surfaces carefully for loose rivets, cracks, or buckling. The interior of the wheel well may give further indications of excessive gust conditions. Inspect the top and bottom fuselage skin. An excessive bending moment may have left wrinkles of a diagonal nature in these areas.

Inspect the surface of the empennage for wrinkles, buckling, or sheared attachments. Also, inspect the area of attachment of the empennage to the fuselage. These inspections cover

ATA iSpec 2200 Systems Sample

Systems	Subsystems	Title	Systems	Subsystems	Title
21		AIR CONDITIONING	42		INTEGRATED MODULAR AVIONICS
21	-00	General	51		STANDARD PRACTICES AND STRUCTURES - GENERAL
21	-10	Compression	52		DOORS
21	-20	Distribution	53		FUSELAGE
21	-30	Pressurization Control	54		NACELLES/PYLONS
21	-40	Heating	55		STABILIZERS
21	-50	Cooling	56		WINDOWS
21	-60	Temperature Control	57		WINGS
21	-70	Moisture/Air Contaminate Control	60		STANDARD PRACTICES - PROPELLER/ROTOR
22		AUTO FLIGHT	61		PROPELLERS/PROPULSION
23		COMMUNICATIONS	62		ROTOR(S)
24		ELECTRICAL POWER	71		POWER PLANT
25		EQUIPMENT/FURNISHINGS	72		ENGINE TURBINE/TURBOPROP DUCTED FAN/UNDUCTED FAN
26		FIRE PROTECTION	72	-20	AIR INLET SECTION
27		FLIGHT CONTROLS	73		ENGINE FUEL AND CONTROL
28		FUEL	74		IGNITION
29		HYDRAULIC POWER	75		AIR
30		ICE AND RAIN PROTECTION	76		ENGINE CONTROLS
31		INDICATING/RECORDING SYSTEMS	77		ENGINE INDICATING
32		LANDING GEAR	78		EXHAUST
33		LIGHTS	79		OIL
34		NAVIGATION	80		STARTING
35		OXYGEN	81		TURBINES
36		PNEUMATIC	82		WATER INJECTION
37		VACUUM	83		ACCESSORY GEAR-BOXES
38		WATER/WASTE			

This figure shows a representative number of systems/subsystems for demonstration purposes only. It is not all-encompassing and should be viewed as a learning aid to understand the numbering method of the ATA iSpec 2200 System. Consult the specific aircraft maintenance manuals or Airlines for America (A4A) for a complete description of the systems and subsystems.

Figure 10-4. *ATA iSpec 2200 Systems.*

the critical areas. If excessive damage is noted in any of the areas mentioned, the inspection must be continued until all damage is detected.

Lightning Strike

Although lightning strikes to aircraft are extremely rare, if a strike has occurred, the aircraft is carefully inspected to determine the extent of any damage that might have occurred. When lightning strikes an aircraft, the electrical current must be conducted through the structure and be allowed to discharge or dissipate at controlled locations. These controlled locations are primarily the aircraft's static discharge wicks, or on more sophisticated aircraft, null field dischargers. When surges of high-voltage electricity pass through good electrical conductors, such as aluminum or steel, damage is likely to be minimal or nonexistent. When surges of high-voltage electricity pass through non-metallic structures, such as a fiberglass radome, engine cowl or fairing, glass or plastic window, or a composite structure that does not have built-in electrical bonding, burning and more serious damage to the structure could occur. Visual inspection of the structure is required. Look for evidence of degradation, burning, or erosion of the composite resin at all affected structures, electrical bonding straps, static discharge wicks, and null field dischargers.

Bird Strike

When the aircraft is hit by birds during flight, the external areas of the airplane are inspected in the general area of the bird strike. If the initial inspection shows structural damage, then the internal structure of the airplane must be inspected as well. Also, inspect the hydraulic, pneumatic, and any other systems in the area of the bird strike.

Fire Damage

Inspection of aircraft structures that have been subjected to fire or intense heat can be relatively simple if visible damage is present. Visible damage requires repair or replacement. If there is no visible damage, the structural integrity of an aircraft may still have been compromised. Since most structural metallic components of an aircraft have undergone some sort of heat-treatment process during manufacture, an exposure to high heat not encountered during normal operations could severely degrade the design strength of the structure. The strength and airworthiness of an aluminum structure that passes a visual inspection, but is still suspect, can be further determined by use of a conductivity tester. This is a device that uses eddy current and is discussed later in this chapter. Since strength of metals is related to hardness, possible damage to steel structures might be determined by use of a hardness tester, such as a Rockwell C hardness tester. *[Figure 10-5]*

Flood Damage

Like aircraft damaged by fire, aircraft damaged by water can range from minor to severe. This depends on the level of the flood water, whether it was fresh or salt water, and the elapsed time between the flood occurrence and when repairs were initiated. Any parts that were totally submerged are completely disassembled, thoroughly cleaned, dried, and treated with a corrosion inhibitor. Many parts might have to be replaced, particularly interior carpeting, seats, side panels, and instruments. Since water serves as an electrolyte that promotes corrosion, all traces of water and salt must be removed before the aircraft can again be considered airworthy.

Seaplanes

Because they operate in an environment that accelerates corrosion, seaplanes must be carefully inspected for corrosion and conditions that promote corrosion. Inspect bilge areas for waste hydraulic fluids, water, dirt, drill chips, and other debris. Additionally, since seaplanes often encounter excessive stress from the pounding of rough water at high speeds, inspect for loose rivets and other fasteners; stretched, bent or cracked skins; damage to the float attach fitting; and general wear and tear on the entire structure.

Aerial Application Aircraft

Two primary factors that make inspecting these aircraft different from other aircraft are the corrosive nature of some of the chemicals used and the typical flight profile. Damaging effects of corrosion may be detected in a much shorter period of time than normal use aircraft. Chemicals may soften the fabric or loosen the fabric tapes of fabric-covered aircraft. Metal aircraft may need to have the paint stripped, cleaned, and repainted and corrosion treated annually. Leading edges of wings and other areas may require protective coatings or tapes. Hardware may require more frequent replacement.

During peak use, these aircraft may fly up to 50 cycles (takeoffs and landings) or more in a day, most likely from an unimproved or grass runway. This can greatly accelerate the failure of normal fatigue items. Landing gear and related items require frequent inspections. Because these aircraft operate almost continuously at very low altitudes, air filters tend to become obstructed more rapidly.

Special Flight Permits

For an aircraft that does not currently meet airworthiness requirements because of an overdue inspection, damage, expired replacement times for time-limited parts, or other reasons, but is capable of safe flight, a special flight permit may be issued. Special flight permits, often referred to as ferry permits, are issued for the following purposes:

- Flying the aircraft to a base where repairs, alterations, or maintenance are to be performed or to a point of storage
- Delivering or exporting the aircraft
- Production flight testing new production aircraft
- Evacuating aircraft from areas of impending danger
- Conducting customer demonstration flights in new production aircraft that have satisfactorily completed production flight tests

Additional information about special flight permits may be found in 14 CFR part 21. Application forms for special flight permits may be requested from the nearest FAA FSDO.

Nondestructive Inspection/Testing

The preceding information in this chapter provided general details regarding aircraft inspection. The remainder of this chapter deals with several methods often used on specific components or areas on an aircraft when carrying out the more specific inspections. They are referred to as nondestructive inspection (NDI) or nondestructive testing (NDT). The objective of NDI and NDT is to determine the airworthiness of a component, without damaging it, that would render it unairworthy. Some of these methods are simple, requiring little additional expertise, while others are highly sophisticated and require that the technician be highly trained and specially certified.

Figure 10-5. *Rockwell C Hardness Tester.*

Training, Qualification, & Certification

The product manufacturer or the FAA generally specifies the particular NDI method and procedure to be used in inspection. These NDI requirements are specified in the manufacturer's inspection, maintenance, or overhaul manual, FAA ADs, supplemental structural inspection documents (SSID), or SBs.

The success of any NDI method and procedure depends upon the knowledge, skill, and experience of the NDI personnel involved. The person(s) responsible for detecting and interpreting indications, such as eddy current, x-ray, or ultrasonic NDI, must be qualified and certified to specific FAA or other acceptable government or industry standards, such as MIL-STD-410, Nondestructive Testing Personnel Qualification and Certification or ATA iSpec 105, Guidelines for Training and Qualifying Personnel in Nondestructive Testing Methods. The person must be familiar with the test method, know the potential types of discontinuities peculiar to the material, and be familiar with their effect on the structural integrity of the part. Additional information on NDI may be found by referring to Chapter 5 of FAA AC 43.13-1, Acceptable Methods, Techniques, and Practices—Aircraft Inspection and Repair.

Advantages & Disadvantages of NDI Methods

Figure 10-6 provides a table of the advantages and disadvantages of common NDI methods. This table could be used as a guide for evaluating the most appropriate NDI method when the manufacturer or the FAA has not specified a particular NDI method to be used.

General Techniques

Before conducting NDI, it is necessary to follow preparatory steps in accordance with procedures specific to that type of inspection. Generally, the parts or areas must be thoroughly cleaned. Some parts must be removed from the aircraft or engine. Others might need to have any paint or protective coating stripped. A complete knowledge of the equipment and procedures is essential and, if required, calibration and inspection of the equipment must be current.

Visual Inspection

Visual inspection can be enhanced by looking at the suspect area with a bright light, a magnifying glass, and a mirror. Some defects might be so obvious that further inspection methods are not required. The lack of visible defects does not necessarily mean further inspection is unnecessary. Some defects may lie beneath the surface or may be so small that the human eye, even with the assistance of a magnifying glass, cannot detect them.

Surface Cracks

When searching for surface cracks with a flashlight, direct the light beam at a 5 to 45 degree angle to the inspection surface towards the face. *[Figure 10-7]* Do not direct the light beam at such an angle that the reflected light beam shines directly into the eyes. Keep the eyes above the reflected light beam during the inspection. Determine the extent of any cracks found by directing the light beam at right angles to the crack and tracing its length. Use a 10-power magnifying glass to confirm the existence of a suspected crack. If this is not adequate, use other NDI techniques, such as penetrant, magnetic particle, or eddy current to verify cracks.

Borescope

Inspection by use of a borescope is essentially a visual inspection. A borescope is a device that enables the inspector to see inside areas that could not otherwise be inspected without disassembly. Borescopes are used in aircraft and engine maintenance programs to reduce or eliminate the need for costly teardowns. Aircraft turbine engines have access ports that are specifically designed for borescopes. Borescopes are also used extensively in a variety of aviation maintenance programs to determine the airworthiness of difficult to reach components. Borescopes typically are used to inspect interiors of hydraulic cylinders and valves for pitting, scoring, porosity, and tool marks; search for cracked cylinders in aircraft reciprocating engines; inspect turbojet engine turbine blades and combustion cans; verify the proper placement and fit of seals, bonds, gaskets, and subassemblies in difficult to reach areas; and assess foreign object damage (FOD) in aircraft, airframe, and powerplants. Borescopes may also be used to locate and retrieve foreign objects in engines and airframes.

Borescopes are available in two basic configurations. The simpler of the two is a rigid type, small diameter telescope with a tiny mirror at the end that enables the user to see around corners. The other type uses fiber optics that enable greater flexibility. *[Figure 10-8]* Many borescopes provide images that can be displayed on a computer or video monitor for better interpretation of what is being viewed and to record images for future reference. Most borescopes also include a light to illuminate the area being viewed.

Liquid Penetrant Inspection

Penetrant inspection is a nondestructive test for defects open to the surface in parts made of any nonporous material. It is used with equal success on such metals as aluminum, magnesium, brass, copper, cast iron, stainless steel, and titanium. It may also be used on ceramics, plastics, molded rubber, and glass.

Penetrant inspection detects defects, such as surface cracks or porosity. These defects may be caused by fatigue cracks, shrinkage cracks, shrinkage porosity, cold shuts, grinding and heat-treat cracks, seams, forging laps, and bursts. Penetrant inspection also indicates a lack of bond between joined metals. The main disadvantage of penetrant inspection is that the defect must be open to the surface in order to let the penetrant get into the defect. For this reason, if the part in question is made of material that is magnetic, the use of magnetic particle inspection is generally recommended.

Penetrant inspection uses a penetrating liquid that enters a surface opening and remains there, making it clearly visible to the inspector. It calls for visual examination of the part after it has been processed, increasing the visibility of the defect so that it can be detected. Visibility of the penetrating material is increased by the addition of one or two types of dye: visible or fluorescent.

The visible penetrant kit consists of dye penetrant, dye remover emulsifier, and developer. The fluorescent penetrant inspection kit contains a black light assembly, as well as spray cans of penetrant, cleaner, and developer. The light assembly consists of a power transformer, a flexible power cable, and a hand-held lamp. Due to its size, the lamp may be used in almost any position or location.

The steps for performing a penetrant inspection are:

1. Clean the metal surface thoroughly.
2. Apply penetrant.
3. Remove penetrant with remover emulsifier or cleaner.
4. Dry the part.
5. Apply the developer.
6. Inspect and interpret results.

Interpretation of Results

The success and reliability of a penetrant inspection depends upon the thoroughness that the part was prepared with. Several basic principles applying to penetrant inspection are:

1. The penetrant must enter the defect in order to form an indication. It is important to allow sufficient time so the penetrant can fill the defect. The defect must be clean and free of contaminating materials so that the penetrant is free to enter.
2. If all penetrant is washed out of a defect, an indication cannot be formed. During the washing or rinsing operation, prior to development, it is possible that the penetrant is removed from within the defect, as well as from the surface.
3. Clean cracks are usually easy to detect. Surface openings that are uncontaminated, regardless of how

Method	Advantages	Disadvantages
Visual	• Inexpensive • Highly portable • Immediate results • Minimum training • Minimum part preparation	• Surface discontinuities only • Generally only large discontinuities • Misinterpretation of scratches
Penetrant Dye	• Portable • Inexpensive • Sensitive to very small discontinuities • 30 minutes or less to accomplish • Minimum skill required	• Locate surface defects only • Rough or porous surfaces interfere with test • Part preparation required (removal of finishes and sealant, etc.) • High degree of cleanliness required • Direct visual detection on results required
Magnetic Particle	• Can be portable • Inexpensive • Sensitive to small discontinuities • Immediate results • Moderate skill required • Detects surface and subsurface discontinuities • Relatively fast	• Surface must be accessible • Rough surfaces interfere with test • Part preparation required (removal of finishes and sealant, etc.) • Semi-directional requiring general orientation of field to discontinuity • Ferro-magnetic materials only • Part must be demagnetized after test
Eddy Current	• Portable • Detects surface and subsurface discontinuities • Moderate speed • Immediate results • Sensitive to small discontinuities • Thickness sensitive • Can detect many variables	• Surface must be accessible to probe • Rough surfaces interfere with test • Electrically conductive materials • Skill and training required • Time consuming for large areas
Ultrasonic	• Portable • Inexpensive • Sensitive to very small discontinuities • Immediate results • Little part preparation • Wide range of materials and thickness can be inspected	• Surface must be accessible to probe • Rough surfaces interfere with test • Highly sensitive to sound beam discontinuity orientation • High degree of skill and experience required for exposure and interpretation • Depth of discontinuity not indicated
X-Ray Radiography	• Detects surface and internal flaws • Can inspect hidden areas • Permanent test record obtained • Minimum part preparation	• Safety hazard • Very expensive (slow process) • Highly directional, sensitive to flaw orientation • High degree of skill and experience required for exposure and interpretation • Depth of discontinuity not indicated
Isotope Radiography	• Portable • Less inexpensive than x-ray • Detects surface and internal flaws • Can inspect hidden areas • Permanent test record obtained • Minimum part preparation	• Safety hazard • Must conform to federal and state regulations for handling and use • Highly directional, sensitive to flaw orientation • High degree of skill and experience required for exposure and interpretation • Depth of discontinuity not indicated

Figure 10-6. *Advantages and disadvantages of NDI methods.*

fine, are seldom difficult to detect with the penetrant inspection.

4. The smaller the defect, the longer the penetrating time. Fine crack-like apertures require a longer penetrating time than defects such as pores.
5. When the part to be inspected is made of a material susceptible to magnetism, it should be inspected by a magnetic particle inspection method if the equipment is available.
6. Visible penetrant-type developer, when applied to the surface of a part, dries to a smooth, white coating. As the developer dries, bright red indications appear where there are surface defects. If no red indications appear, there are no surface defects.
7. When conducting the fluorescent penetrant-type inspection, the defects show up (under black light) as a brilliant yellow-green color and the sound areas appear deep blue-violet.
8. It is possible to examine an indication of a defect and to determine its cause as well as its extent. Such an appraisal can be made if something is known about the manufacturing processes that the part has been subjected to.

The size of the indication, or accumulation of penetrant, shows the extent of the defect and the brilliance is a measure of its depth. Deep cracks hold more penetrant and are

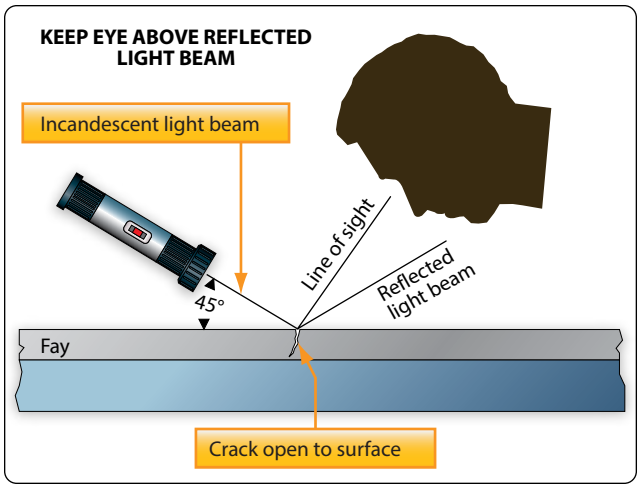

Figure 10-7. *Using a flashlight to inspect for cracks.*

broader and more brilliant. Very fine openings can hold only small amounts of penetrants and appear as fine lines. *[Figure 10-9]*

False Indications

With the penetrant inspection, there are no false indications in the sense that they occur in the magnetic particle inspection. There are, however, two conditions that may create accumulations of penetrant that are sometimes confused with true surface cracks and discontinuities.

The first condition involves indications caused by poor washing. If all the surface penetrant is not removed in the washing or rinsing operation following the penetrant dwell time, the unremoved penetrant is visible. Evidences of incomplete washing are usually easy to identify since the penetrant is in broad areas rather than in the sharp patterns found with true indications. When accumulations of unwashed penetrant are found on a part, the part must be completely reprocessed. Degreasing is recommended for removal of all traces of the penetrant.

False indications may also be created where parts press fit to each other. If a wheel is press fit onto a shaft, penetrant shows an indication at the fit line. This is perfectly normal since the two parts are not meant to be welded together. Indications of this type are easy to identify since they are regular in form and shape.

Eddy Current Inspection

Electromagnetic analysis is a term describing the broad spectrum of electronic test methods involving the intersection of magnetic fields and circulatory currents. The most widely used technique is the eddy current. Eddy currents are composed of free electrons under the influence of an induced electromagnetic field that are made to "drift" through metal.

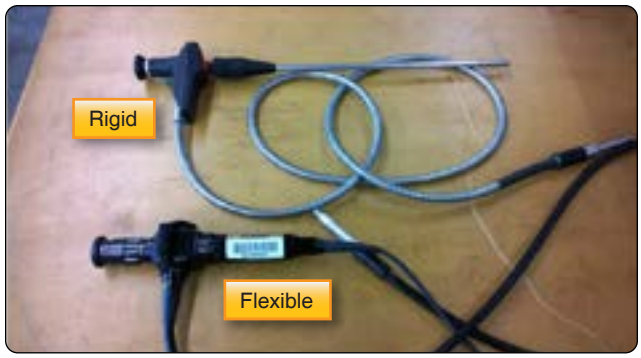

Figure 10-8. *Rigid and flexible borescopes.*

Eddy current is used to detect surface cracks, pits, subsurface cracks, corrosion on inner surfaces, and to determine alloy and heat-treat condition.

Eddy current is used in aircraft maintenance to inspect jet engine turbine shafts and vanes, wing skins, wheels, bolt holes, and spark plug bores for cracks, heat, or frame damage. Eddy current may also be used in repair of aluminum aircraft damaged by fire or excessive heat. Different meter readings are seen when the same metal is in different hardness states. Readings in the affected area are compared with identical materials in known unaffected areas for comparison. A difference in readings indicates a difference in the hardness state of the affected area. In aircraft manufacturing plants, eddy current is used to inspect castings, stampings, machine parts, forgings, and extrusions. *Figure 10-10* shows a technician performing an eddy current inspection on a fan blade.

Basic Principles

When an alternating current (AC) is passed through a coil, it develops a magnetic field around the coil, which in turn induces a voltage of opposite polarity in the coil and opposes the flow of original current. If this coil is placed in such a way that the magnetic field passes through an electrically conducting specimen, eddy currents are induced into the specimen. The eddy currents create their own field that varies the original field's opposition to the flow of original current. The specimen's susceptibility to eddy currents determines the current flow through the coil.

The magnitude and phase of this counter field is dependent primarily upon the resistance and permeability of the specimen under consideration and enables us to make a qualitative determination of various physical properties of the test material. The interaction of the eddy current field with the original field results is a power change that can be measured by utilizing electronic circuitry similar to a Wheatstone bridge.

Figure 10-9. *Dye penetrant inspection.*

Principles of Operations

Eddy currents are induced in a test article when an AC is applied to a test coil (probe). The AC in the coil induces an alternating magnetic field in the article, causing eddy currents to flow in the article. *[Figure 10-11]*

Flaws in or thickness changes of the test-piece influence the flow of eddy currents and change the impedance of the coil accordingly. *[Figure 10-12]* Instruments display the impedance changes either by impedance plane plots or by needle deflection. *[Figure 10-13]*

The specimen is either placed in or passed through the field of an electromagnetic induction coil, and its effect on the impedance of the coil or on the voltage output of one or more test coils is observed. The process that involves electric fields made to explore a test piece for various conditions involves the transmission of energy through the specimen much like the transmission of x-rays, heat, or ultrasound.

Eddy current inspection can frequently be performed without removing the surface coatings, such as primer, paint, and anodized films. It can be effective in detecting surface and subsurface corrosion, pots, and heat-treat condition.

Eddy Current Instruments

A wide variety of eddy current test instruments are available. The eddy current test instrument performs three basic functions: generating, receiving, and displaying. The generating portion of the unit provides an alternating current to the test coil. The receiving section processes the signal from the test coil to the required form and amplitude for display. Instrument outputs or displays consist of a variety of visual, audible, storage, or transfer techniques utilizing meters, video displays, chart recorders, alarms, magnetic tape, computers, and electrical or electronic relays.

A reference standard is required for the calibration of eddy current test equipment. A reference standard is made from the same material as the item is to be tested. A reference standard contains known flaws or cracks and could include items, such as a flat surface notch, a fastener head, a fastener hole, or a countersink hole. *Figures 10-14, 10-15,* and *10-16* show typical surface cracks, subsurface cracks, and structural corrosion that can be detected with eddy current techniques.

Ultrasonic Inspection

Ultrasonic inspection is an NDI technique that uses sound energy moving through the test specimen to detect flaws. The sound energy passing through the specimen is displayed on a cathode ray tube (CRT), a liquid crystal display (LCD) computer data program, or video/camera medium. Indications of the front and back surface and internal/external conditions appear as vertical signals on the CRT screen or nodes of data in the computer test program. *[Figure 10-17]* There are three types of display patterns: “A” scan, “B” scan, and “C” scan. Each scan provides a different picture or view of the specimen being tested. *[Figure 10-18]*

Ultrasonic detection equipment makes it possible to locate defects in all types of materials. Minute cracks, checks, and voids too small to be seen by x-ray can be located by ultrasonic inspection. An ultrasonic test instrument requires access to only one surface of the material to be inspected and can be used with either straight line or angle beam testing techniques.

Two basic methods are used for ultrasonic inspection. The first of these methods is immersion testing. In this method of inspection, the part under examination and the search unit are completely immersed in a liquid couplant, such as water or other suitable fluids.

The second method is called contact testing. It is readily adapted to field use and is the method discussed in this chapter. In this method, the part under examination and the search unit are coupled with a viscous material, liquid, or a paste that wets both the face of the search unit and the material under examination.

Figure 10-10. *Eddy current inspection.*

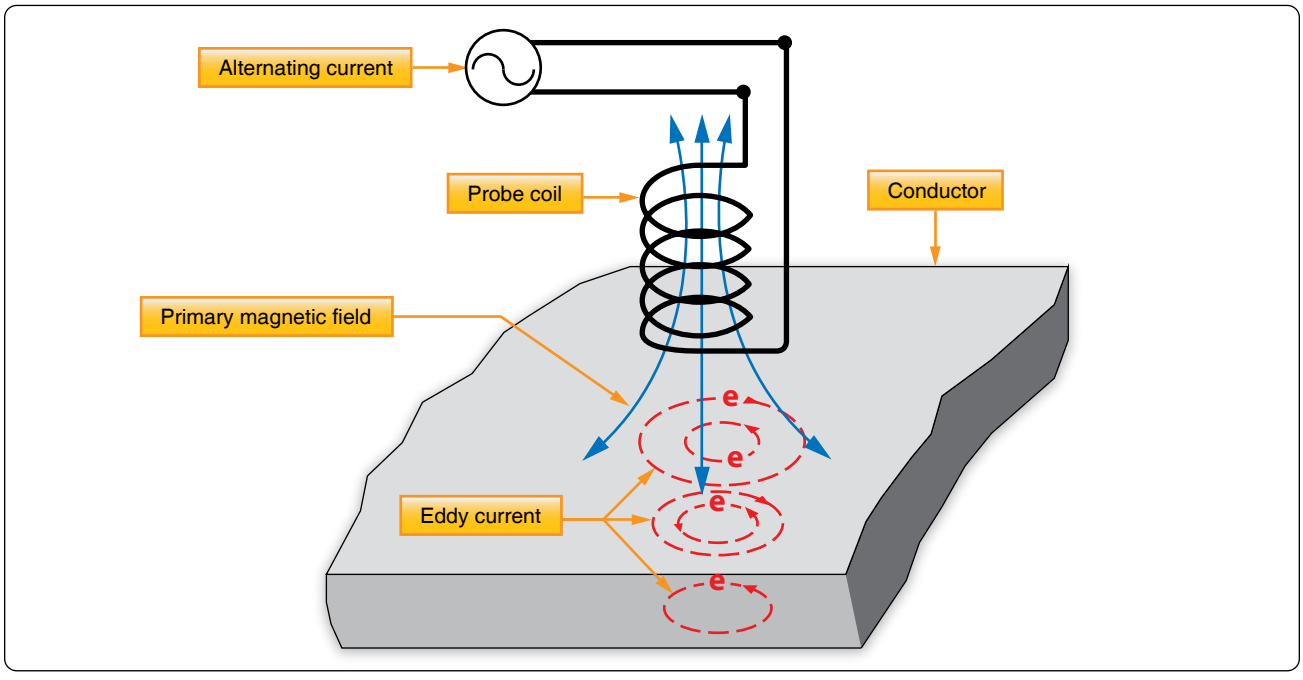

Figure 10-11. *Generating an eddy current.*

There are three basic ultrasonic inspection methods: pulse echo, through-transmission, and resonance. Through-transmission and pulse echo are shown in *Figure 10-19*.

Pulse Echo

Flaws are detected by measuring the amplitude of signals reflected and the time required for these signals to travel between specific surfaces and the discontinuity. *[Figure 10-20]*

The time base, triggered simultaneously with each transmission pulse, causes a spot to sweep across the screen of the CRT or LCD. The spot sweeps from left to right across the face of the scope 50 to 5,000 times per second or higher if required for high-speed automated scanning. Due to the speed of the cycle of transmitting and receiving, the picture on the oscilloscope appears to be stationary.

A few microseconds after the sweep is initiated, the rate generator electrically excites the pulser, and the pulser in turn emits an electrical pulse. The transducer converts this pulse into a short train of ultrasonic sound waves. If the interfaces

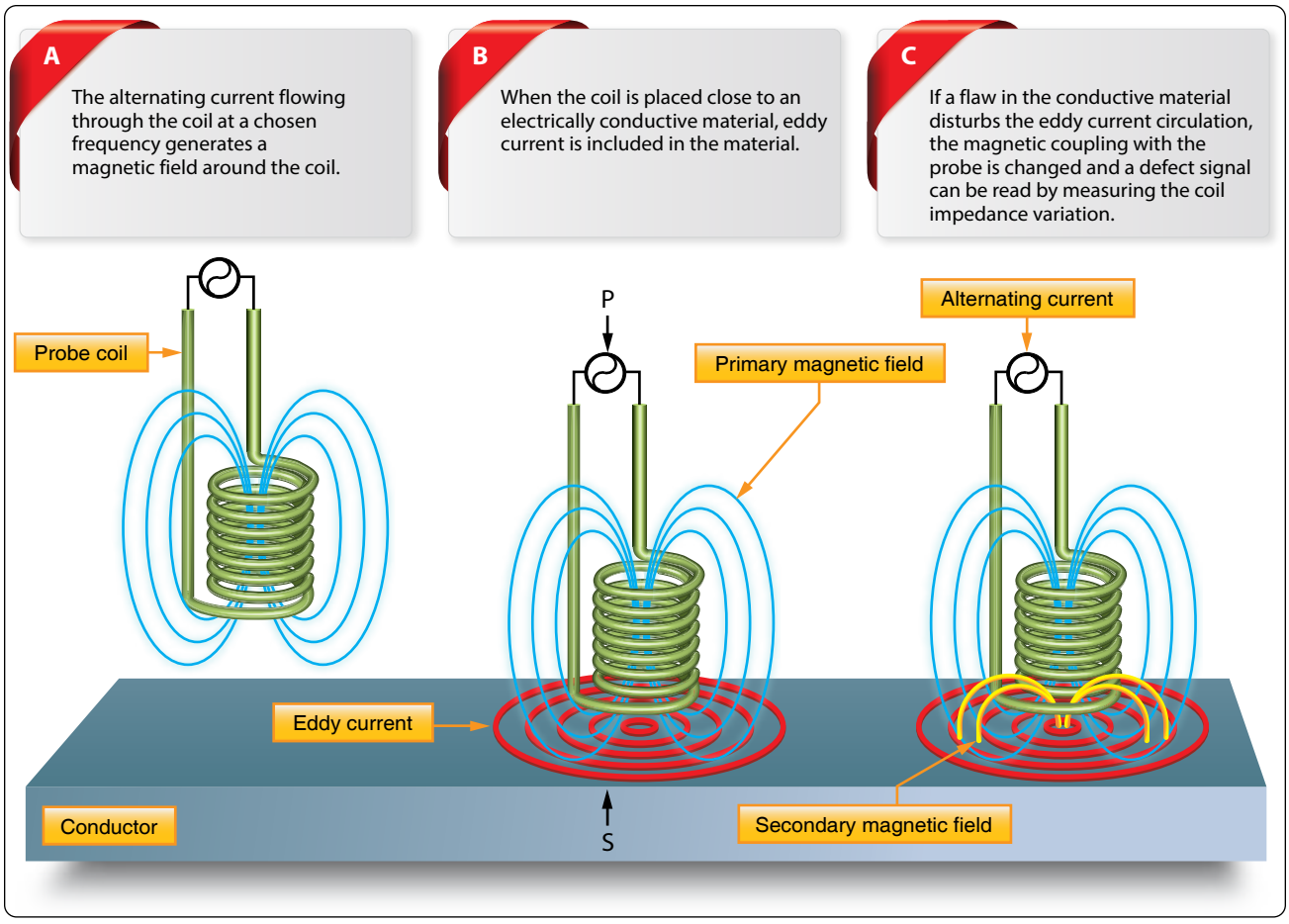

Figure 10-12. *Detecting an eddy current.*

Figure 10-13. *Impedance plane test.*

of the transducer and the specimen are properly oriented, the ultrasound is reflected back to the transducer when it reaches the internal flaw and the opposite surface of the specimen. The time interval between the transmission of the initial impulse and the reception of the signals from within the specimen are measured by the timing circuits. The reflected pulse received by the transducer is amplified, transmitted to, and displayed on the instrument screen. The pulse is displayed in the same relationship to the front and back pulses as the flaw is in relation to the front and back surfaces of the specimen. *[Figure 10-21]*

Pulse-echo instruments may also be used to detect flaws not directly underneath the probe by use of the angle beam testing method. Angle beam testing differs from straight beam testing only in the manner that the ultrasonic waves pass through the material being tested. As shown in *Figure 10-22*, the beam is projected into the material at an acute angle to the surface by means of a crystal cut at an angle and mounted in plastic. The beam, or a portion thereof, reflects successively from the surfaces of the material or any other discontinuity, including the edge of the piece. In straight beam testing, the horizontal distance on the screen between the initial pulse and the first back reflection represents the thickness of the piece; while in angle beam testing, this distance represents the width of the material between the searching unit and the opposite edge of the piece.

Through-Transmission

Through-transmission inspection uses two transducers, one to generate the pulse and another placed on the opposite surface

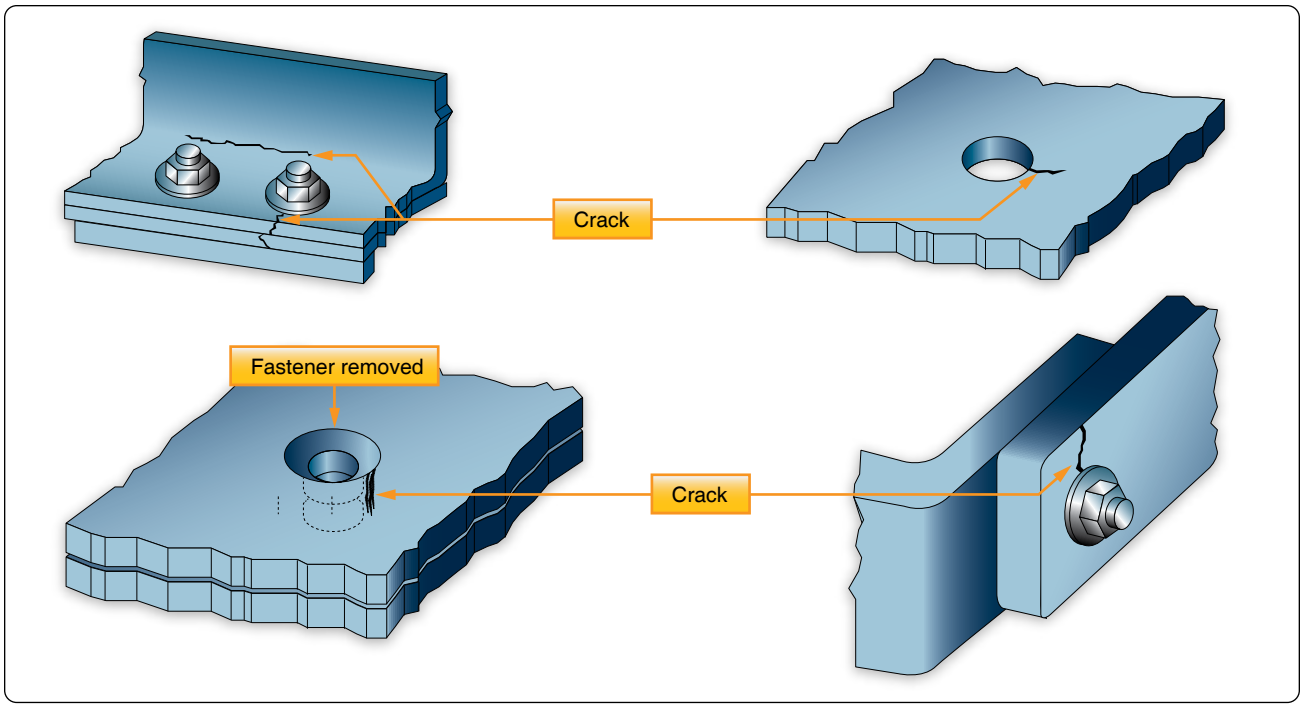

Figure 10-14. *Typical surface cracks.*

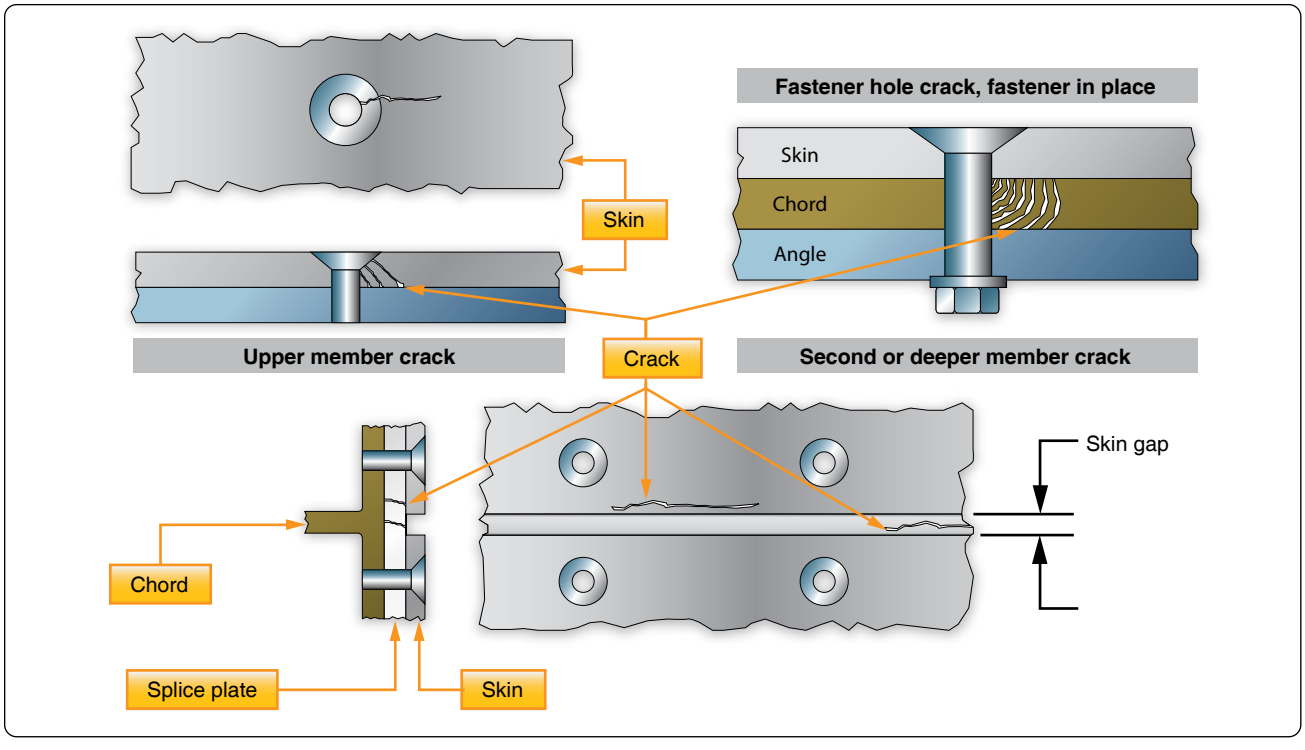

Figure 10-15. *Typical subsurface cracks.*

to receive it. A disruption in the sound path indicates a flaw and is displayed on the instrument screen. Through-transmission is less sensitive to small defects than the pulse-echo method.

Resonance

This system differs from the pulse method in that the frequency of transmission may be continuously varied. The resonance method is used principally for thickness

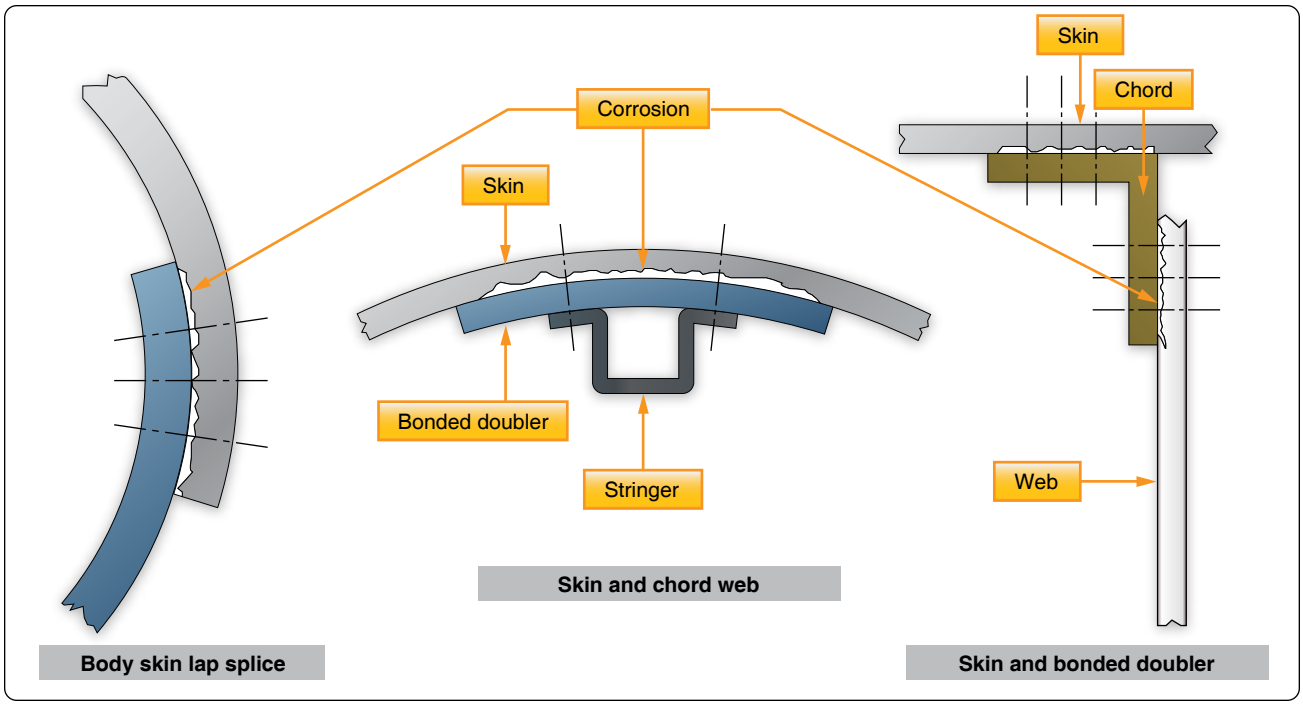

Figure 10-16. *Typical structural corrosion.*

Figure 10-17. *Ultrasonic inspection.*

measurements when the two sides of the material being tested are smooth and parallel and the backside is inaccessible. The point where the frequency matches the resonance point of the material being tested is the thickness determining factor. It is necessary that the frequency of the ultrasonic waves corresponding to a particular dial setting be accurately known. Checks are made with standard test blocks to guard against possible drift of frequency.

If the frequency of an ultrasonic wave is such that its wavelength is twice the thickness of a specimen (fundamental frequency), then the reflected wave arrives back at the transducer in the same phase as the original transmission so that strengthening of the signal occurs. This results from constructive interference or a resonance and is shown as a high amplitude value on the indicating screen. If the frequency is increased such that three times the wavelength equals four times the thickness, the reflected signal returns completely out of phase with the transmitted signal and cancellation occurs. Further increase of the frequency causes the wavelength to be equal to the thickness again and gives a reflected signal in phase with the transmitted signal and a resonance once more. By starting at the fundamental frequency and gradually increasing the frequency, the successive cancellations and resonances can be noted and the readings used to check the fundamental frequency reading. *[Figure 10-23]*

In some instruments, the oscillator circuit contains a motor-driven capacitor that changes the frequency of the oscillator. *[Figure 10-24]* In other instruments, the frequency is changed by electronic means. The change in frequency is synchronized with the horizontal sweep of a CRT. The horizontal axis represents a frequency range. If the frequency range contains resonances, the circuitry is arranged to present these vertically. Calibrated transparent scales are then placed in front of the tube and the thickness can be read directly. The instruments normally operate between 0.25 millicycle (mc) and 10 mc in four or five bands.

The resonance thickness instrument can be used to test the thickness of such metals as steel, cast iron, brass, nickel, copper, silver, lead, aluminum, and magnesium. In addition, areas of corrosion or wear on tanks, tubing, airplane wing skins,

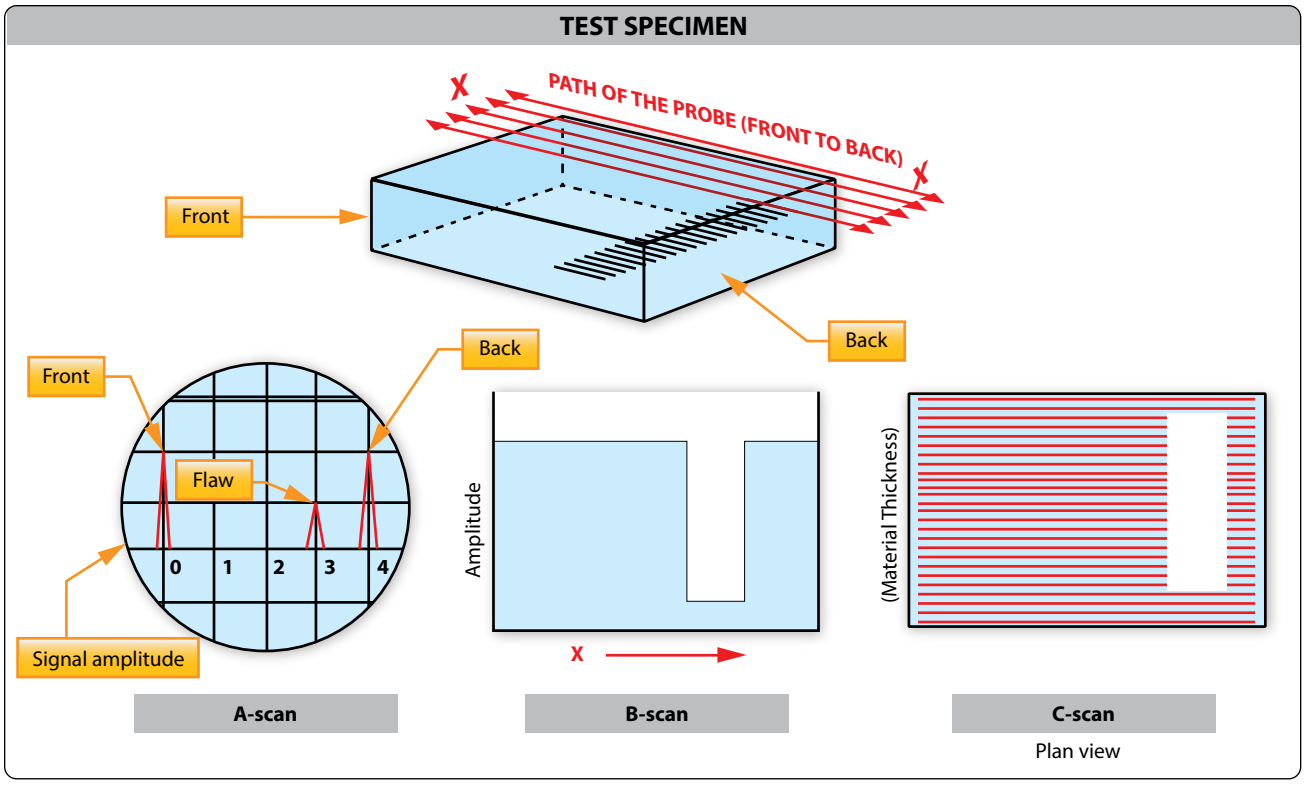

Figure 10-18. *Typical structural corrosion.*

and other structures or products can be located and evaluated. Direct reading dial-operated units are available that measure thickness between 0.025 inch and 3 inches with an accuracy of better than ±1 percent. Ultrasonic inspection requires a skilled operator who is familiar with the equipment being used, as well as the inspection method to be used for the many different parts being tested. *[Figure 10-25]*

Ultrasonic Instruments

A portable, battery-powered ultrasonic instrument is used for field inspection of airplane structure. The instrument generates an ultrasonic pulse, detects and amplifies the returning echo, and displays the detected signal on a CRT or similar display. Piezoelectric transducers produce longitudinal or shear waves, the most commonly used waveforms for aircraft structural inspection.

Reference Standards

Reference standards are used to calibrate the ultrasonic instrument. Reference standards serve two purposes: to provide an ultrasonic response pattern that is related to the part being inspected and to establish the required inspection sensitivity. To obtain a representative response pattern, the reference standard configuration is the same as that of the test structure or is a configuration that provides an ultrasonic response pattern representative of the test structure. The reference standard contains a simulated defect (notch) that is positioned to provide a calibration signal representative of the expected defect. The notch size is chosen to establish inspection sensitivity (response to the expected defect size). The inspection procedure gives a detailed description of the required reference standard.

Couplants

Inspection with ultrasonics is limited to the part in contact with the transducer. A layer of couplant is required to couple the transducer to the test piece, because ultrasonic energy does not travel through air. Some typical couplants used are water, glycerin, motor oils, and grease.

Inspection of Bonded Structures

Ultrasonic inspection is finding increasing application in aircraft bonded construction and repair. Many configurations and types of bonded structures are in use in aircraft. All of these variations complicate the application of ultrasonic inspections. An inspection method that works well on one part or one area of the part may not be applicable for different parts or areas of the same part. Some of the variables in the types of bonded structures are as follows:

- Top skin material is made from different materials and thickness
- Different types and thickness of adhesives are used in bonded structures

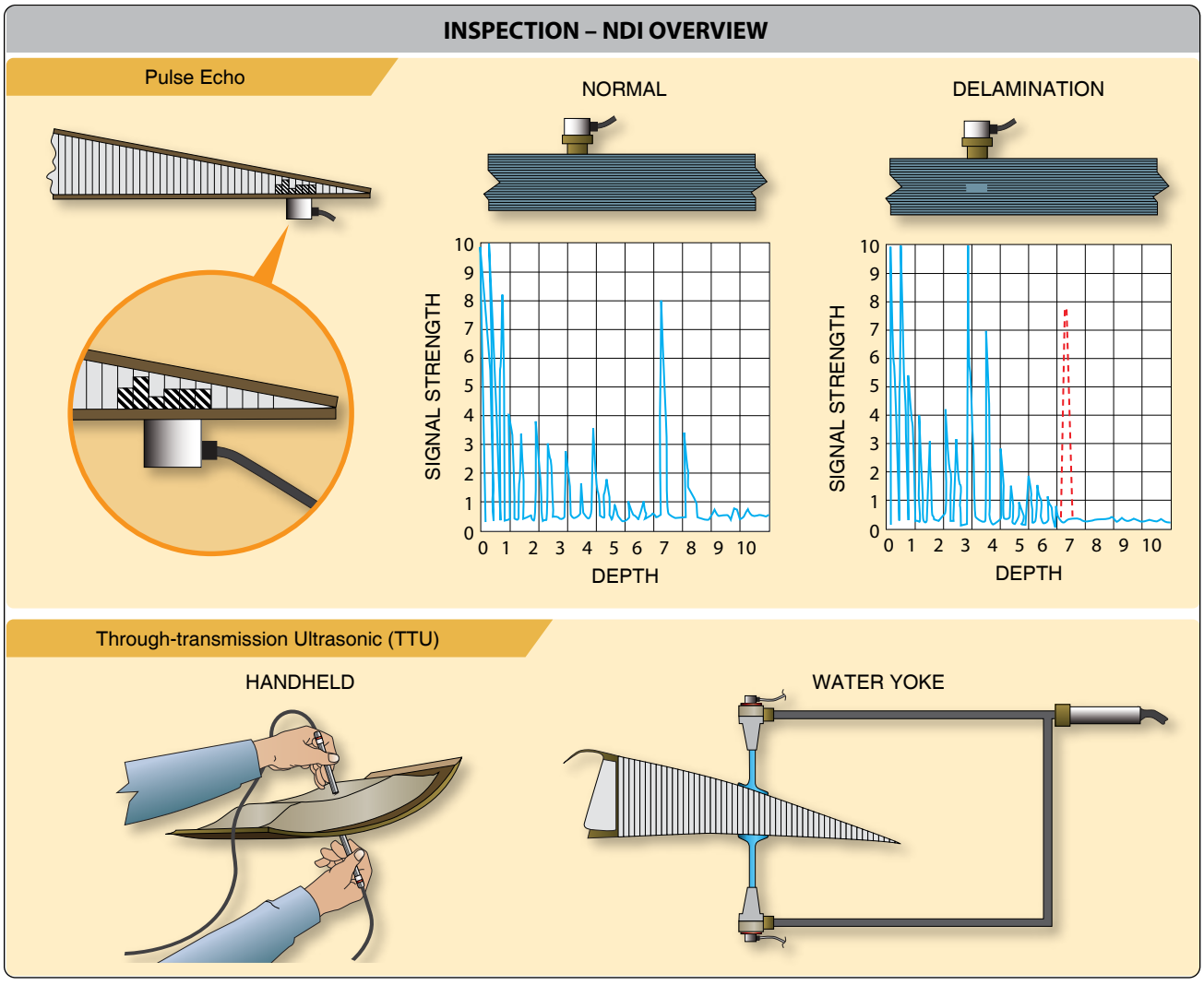

Figure 10-19. *Through-transmission and pulse echo indications.*

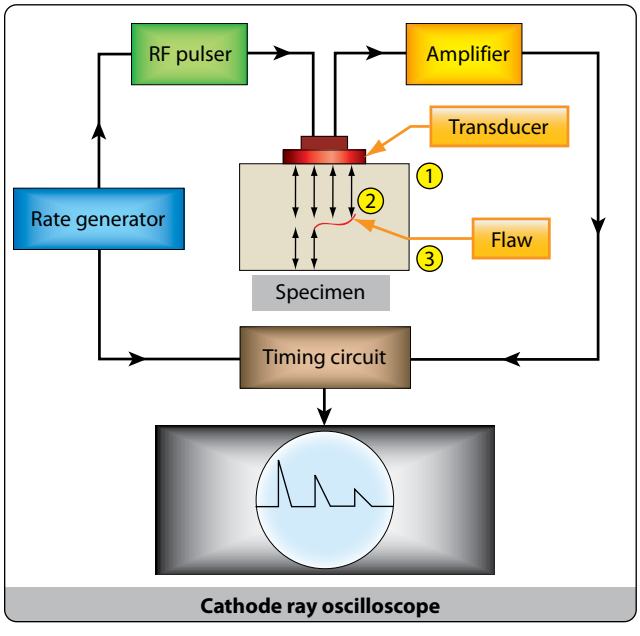

Figure 10-20. *Block diagram of basic pulse-echo system.*

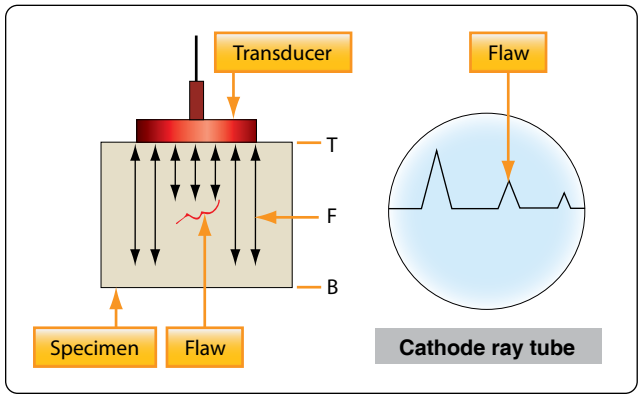

Figure 10-21. *Pulse-echo display in relationship to flaw detection.*

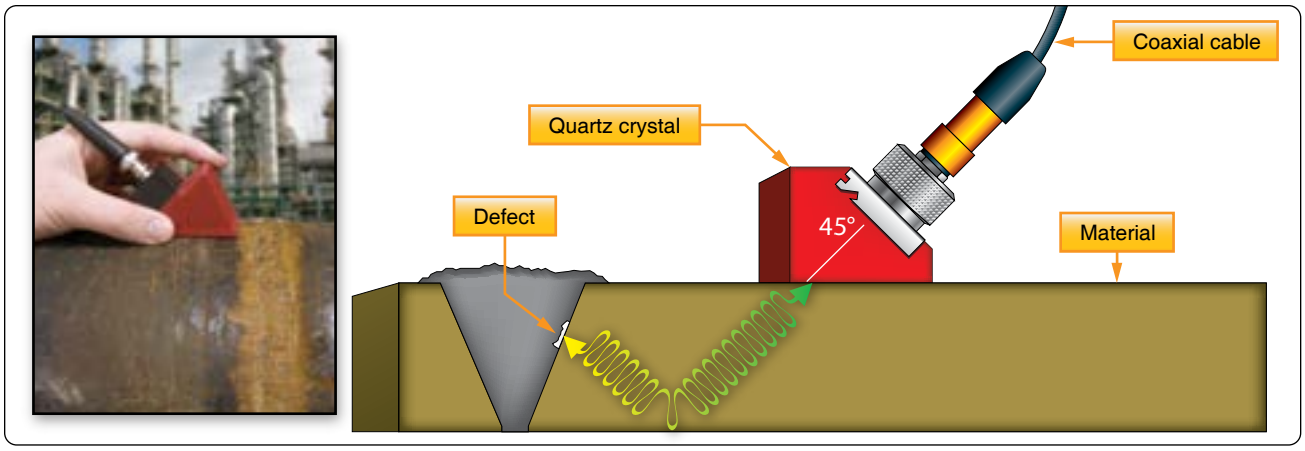

Figure 10-22. *Pulse-echo angle beam testing.*

- Underlying structures contain differences in core material, cell size, thickness, height, back skin material and thickness, doublers (material and thickness), closure member attachments, foam adhesive, steps in skins, internal ribs, and laminates (number of layers, layer thickness, and layer material)
- The top only or top and bottom skin of a bonded structure may be accessible

Types of Defects

Defects can be separated into five general types to represent the various areas of bonded and laminate structures as follows:

1. Type I—disbonds or voids in an outer skin-to-adhesive interface.
2. Type II—disbonds or voids at the adhesive-to-core interface.
3. Type III—voids between layers of a laminate.
4. Type IV—voids in foam adhesive or disbonds between the adhesive and a closure member at core-to-closure member joints.
5. Type V—water in the core.

Acoustic Emission Inspection

Acoustic emission is an NDI technique that involves the placing of acoustic emission sensors at various locations on an aircraft structure and then applying a load or stress. The materials emit sound and stress waves that take the form of ultrasonic pulses. Cracks and areas of corrosion in the stressed airframe structure emit sound waves that are registered by the sensors. These acoustic emission bursts can be used to locate flaws and to evaluate their rate of growth as a function of applied stress. Acoustic emission testing has an advantage over other NDI methods in that it can detect and locate all of the activated flaws in a structure in one test. Because of the complexity of aircraft structures,

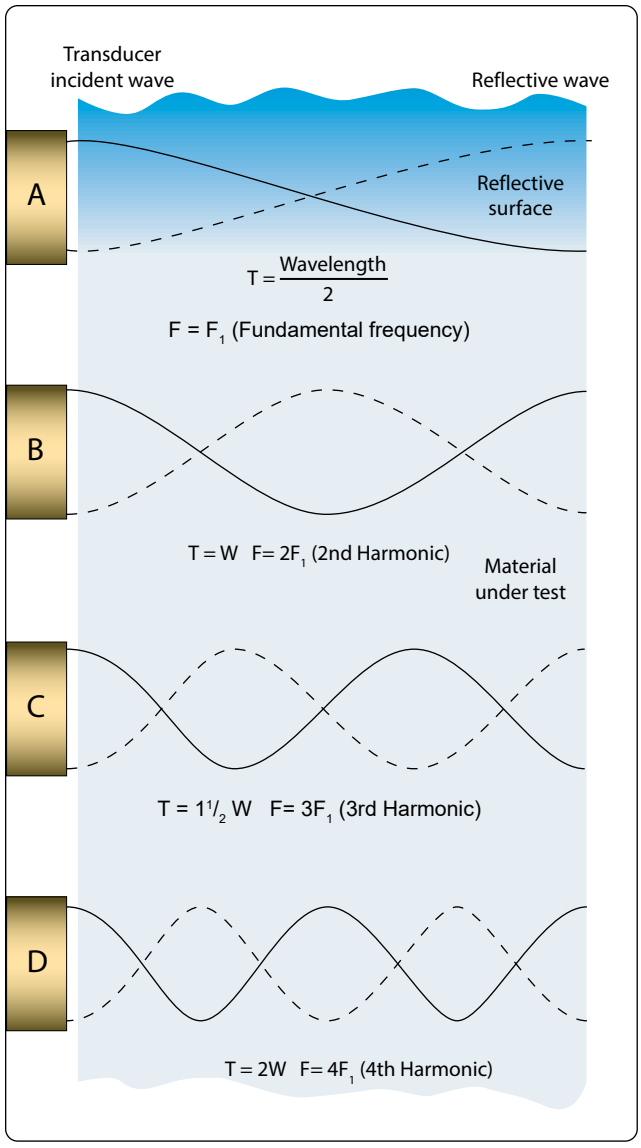

Figure 10-23. *Conditions of ultrasonic resonance in a metal plate.*

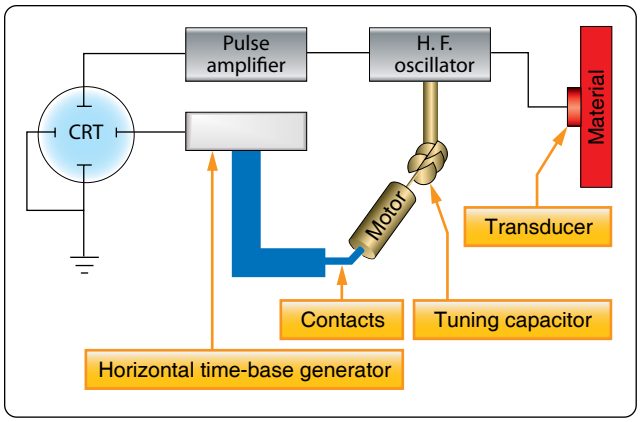

Figure 10-24. *Block diagram of resonance thickness measuring system.*

Figure 10-25. *Ultrasonic inspection of a composite structure.*

application of acoustic emission testing to aircraft has required a new level of sophistication in testing technique and data interpretation.

Magnetic Particle Inspection

Magnetic particle inspection is a method of detecting invisible cracks and other defects in ferromagnetic materials, such as iron and steel. It is not applicable to nonmagnetic materials. In rapidly rotating, reciprocating, vibrating, and other highly-stressed aircraft parts, small defects often develop to the point that they cause complete failure of the part. Magnetic particle inspection has proven extremely reliable for the rapid detection of such defects located on or near the surface. With this method of inspection, the location of the defect is indicated and the approximate size and shape are outlined.

The inspection process consists of magnetizing the part and then applying ferromagnetic particles to the surface area to be inspected. The ferromagnetic particles (indicating medium) may be held in suspension in a liquid that is flushed over the part; the part may be immersed in the suspension liquid; or the particles, in dry powder form, may be dusted over the surface of the part. The wet process is more commonly used in the inspection of aircraft parts.

If a discontinuity is present, the magnetic lines of force are disturbed and opposite poles exist on either side of the discontinuity. The magnetized particles thus form a pattern in the magnetic field between the opposite poles. This pattern, known as an "indication," assumes the approximate shape of the surface projection of the discontinuity. A discontinuity may be defined as an interruption in the normal physical structure or configuration of a part, such as a crack, forging lap, seam, inclusion, porosity, and the like. A discontinuity may or may not affect the usefulness of a part.

Development of Indications

When a discontinuity in a magnetized material is open to the surface and a magnetic substance (indicating medium) is available on the surface, the flux leakage at the discontinuity tends to form the indicating medium into a path of higher permeability. (Permeability is a term used to refer to the ease that a magnetic flux can be established in a given magnetic circuit.) Because of the magnetism in the part and the adherence of the magnetic particles to each other, the indication remains on the surface of the part in the form of an approximate outline of the discontinuity that is immediately below it. The same action takes place when the discontinuity is not open to the surface, but since the amount of flux leakage is less, fewer particles are held in place and a fainter and less sharply defined indication is obtained.

If the discontinuity is very far below the surface, there may be no flux leakage and no indication on the surface. The flux leakage at a transverse discontinuity is shown in *Figure 10-26*. The flux leakage at a longitudinal discontinuity is shown in *Figure 10-27*.

Types of Discontinuities Disclosed

The following types of discontinuities are normally detected by the magnetic particle test: cracks, laps, seams, cold shuts, inclusions, splits, tears, pipes, and voids. All of these may affect the reliability of parts in service.

Cracks, splits, bursts, tears, seams, voids, and pipes are formed by an actual parting or rupture of the solid metal. Cold shuts and laps are folds that have been formed in the metal, interrupting its continuity.

Inclusions are foreign material formed by impurities in the metal during the metal processing stages. They may consist, for example, of bits of furnace lining picked up during the melting of the basic metal or of other foreign constituents. Inclusions interrupt the continuity of the metal, because they

prevent the joining or welding of adjacent faces of the metal.

Preparation of Parts for Testing

Grease, oil, and dirt must be cleaned from all parts before they are tested. Cleaning is very important since any grease or other foreign material present can produce nonrelevant indications due to magnetic particles adhering to the foreign material as the suspension drains from the part.

Grease or foreign material in sufficient amount over a discontinuity may also prevent the formation of a pattern at the discontinuity. It is not advisable to depend upon the magnetic particle suspension to clean the part. Cleaning by suspension is not thorough and any foreign materials so removed from the part contaminates the suspension, thereby reducing its effectiveness.

In the dry procedure, thorough cleaning is absolutely necessary. Grease or other foreign material holds the magnetic powder, resulting in nonrelevant indications and making it impossible to distribute the indicating medium evenly over the part's surface. All small openings and oil holes leading to internal passages or cavities must be plugged with paraffin or other suitable nonabrasive material.

Coatings of cadmium, copper, tin, and zinc do not interfere with the satisfactory performance of magnetic particle inspection, unless the coatings are unusually heavy or the discontinuities to be detected are unusually small.

Chromium and nickel plating generally do not interfere with indications of cracks open to the surface of the base metal, but prevent indications of fine discontinuities, such as inclusions. Because it is more strongly magnetic, nickel plating is more effective than chromium plating in preventing the formation of indications.

Effect of Flux Direction

To locate a defect in a part, it is essential that the magnetic lines of force pass approximately perpendicular to the defect. It is, therefore, necessary to induce magnetic flux in more than one direction, since defects are likely to exist at any angle to the major axis of the part. This requires two separate magnetizing operations, referred to as circular magnetization and longitudinal magnetization. The effect of flux direction is illustrated in *Figure 10-28.*

Circular magnetization is the induction of a magnetic field consisting of concentric circles of force about and within the part. This is achieved by passing electric current through the part, locating defects running approximately parallel to the axis of the part. *Figure 10-29* illustrates circular magnetization of a crankshaft. In longitudinal magnetization, the magnetic field is produced in a direction parallel to the long axis of the part. This is accomplished by placing the part in a solenoid excited by electric current. The metal part then becomes the core of an electromagnet and is magnetized by induction from the magnetic field created in the solenoid. In longitudinal magnetization of long parts, the solenoid must be moved along the part in order to magnetize it. *[Figure 10-30]* This is necessary to ensure adequate field strength throughout the entire length of the part.

Solenoids produce effective magnetization for approximately 12 inches from each end of the coil, thus accommodating parts or sections approximately 30 inches in length. Longitudinal magnetization equivalent to that obtained by a solenoid may be accomplished by wrapping a flexible electrical conductor around the part. Although this method is not as convenient, it has an advantage in that the coils conform more closely to the shape of the part, producing a somewhat more uniform magnetization. The flexible coil method is also useful for large or irregularly-shaped parts when standard solenoids are not available.

Effect of Flux Density

The effectiveness of the magnetic particle inspection also depends on the flux density or field strength at the surface of the part when the indicating medium is applied. As the flux density in the part is increased, the sensitivity of the test increases, because of the greater flux leakages at discontinuities and the resulting improved formation of magnetic particle patterns.

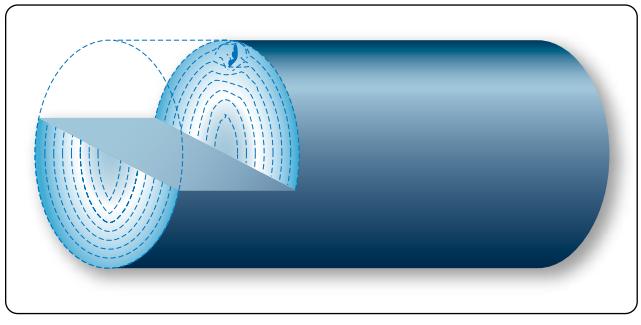

Figure 10-26. *Flux leakage at transverse discontinuity.*

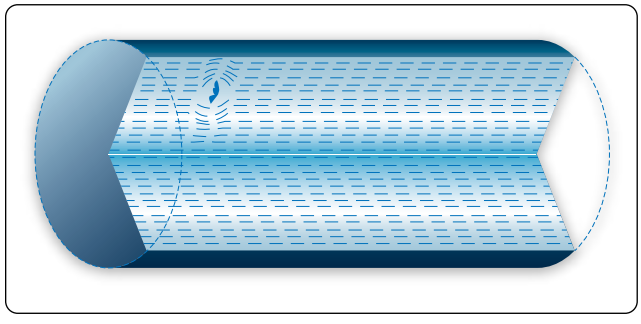

Figure 10-27. *Flux leakage at longitudinal discontinuity.*

Excessively high flux densities may form nonrelevant indications, such as patterns of the grain flow in the material. These indications interfere with the detection of patterns resulting from significant discontinuities. It is therefore necessary to use a field strength high enough to reveal all possible harmful discontinuities, but not strong enough to produce confusing nonrelevant indications.

Magnetizing Methods

When a part is magnetized, the field strength in the part increases to a maximum for the particular magnetizing force and remains at this maximum as long as the magnetizing force is maintained.

When the magnetizing force is removed, the field strength decreases to a lower residual value depending on the magnetic properties of the material and the shape of the part. These magnetic characteristics determine whether the continuous or residual method is used in magnetizing the part.

In the continuous inspection method, the part is magnetized and the indicating medium applied while the magnetizing force is maintained. The available flux density in the part is thus at a maximum. The maximum value of flux depends directly upon the magnetizing force and the permeability of the material that the part is made of.

The continuous method may be used in practically all circular and longitudinal magnetization procedures. The continuous procedure provides greater sensitivity than the residual procedure, particularly in locating subsurface discontinuities. The highly critical nature of aircraft parts and assemblies and the necessity for subsurface inspection in many applications have resulted in the continuous method being more widely used. Since the continuous procedure reveals more nonsignificant discontinuities than the residual procedure, careful and intelligent interpretation and evaluation of discontinuities revealed by this procedure are necessary.

The residual inspection procedure involves magnetization of the part and application of the indicating medium after the magnetizing force has been removed. This procedure relies on the residual or permanent magnetism in the part and is more practical than the continuous procedure when magnetization is accomplished by flexible coils wrapped around the part. In general, the residual procedure is used only with steels that have been heat-treated for stressed applications.

Identification of Indications

The correct evaluation of the character of indications is extremely important but is sometimes difficult to make from observation of the indications alone. The principal distinguishing features of indications are shape, buildup, width, and sharpness of outline. These characteristics are more valuable in distinguishing between types of discontinuities than in determining their severity. Careful observation of the character of the magnetic particle pattern

Figure 10-29. *Circular magnetization of a crankshaft.*

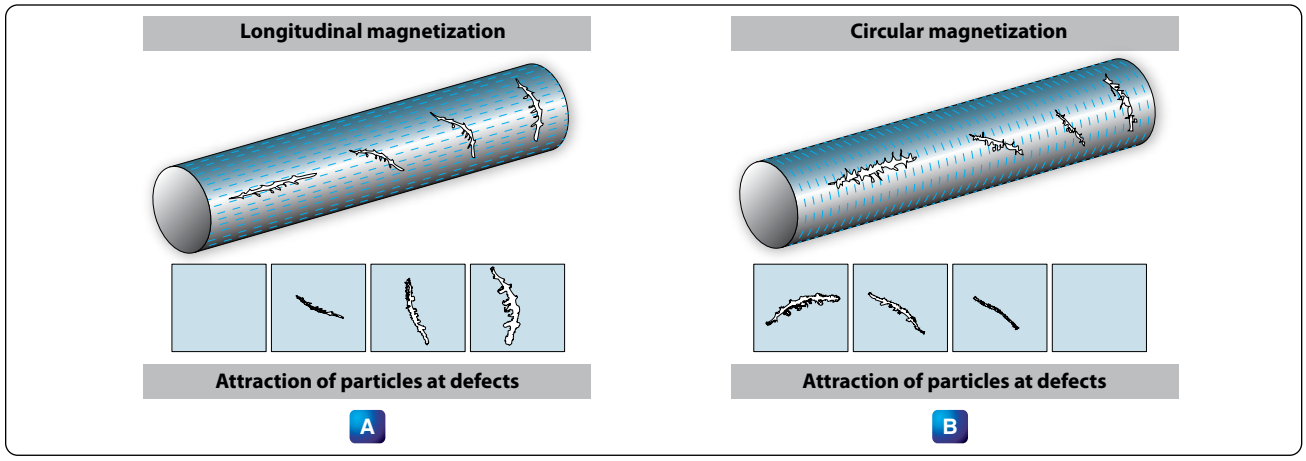

Figure 10-28. *Effect of flux direction on strength of indication.*

must always be included in the complete evaluation of the significance of an indicated discontinuity.

The most readily distinguished indications are those produced by cracks open to the surface. These discontinuities include fatigue cracks, heat-treat cracks, shrink cracks in welds and castings, and grinding cracks. An example of a fatigue crack is shown in *Figure 10-31.*

Magnaglo Inspection

Magnaglo inspection is similar to the preceding method, but differs in that a fluorescent particle solution is used and the inspection is made under black light. *[Figure 10-32]* Efficiency of inspection is increased by the neon-like glow of defects allowing smaller flaw indications to be seen. This is an excellent method for use on gears, threaded parts, and aircraft engine components. The reddish-brown liquid spray or bath that is used consists of Magnaglo paste mixed with a light oil at the ratio of 0.10 to 0.25 ounce of paste per gallon of oil. After inspection, the part must be demagnetized and rinsed with a cleaning solvent.

Magnetizing Equipment

Fixed (Nonportable) General Purpose Unit A fixed, general purpose unit provides direct current (DC) for wet, continuous, or residual magnetization procedures. *[Figure 10-33]* Circular or longitudinal magnetization may be used, and it may be powered with rectified AC, as well as DC. The contact heads provide the electrical terminals for circular magnetization. One head is fixed in position with its contact plate mounted on a shaft surrounded by a pressure spring so that the plate may be moved longitudinally. The plate is maintained in the extended position by the spring until pressure transmitted through the work from the movable head forces it back.

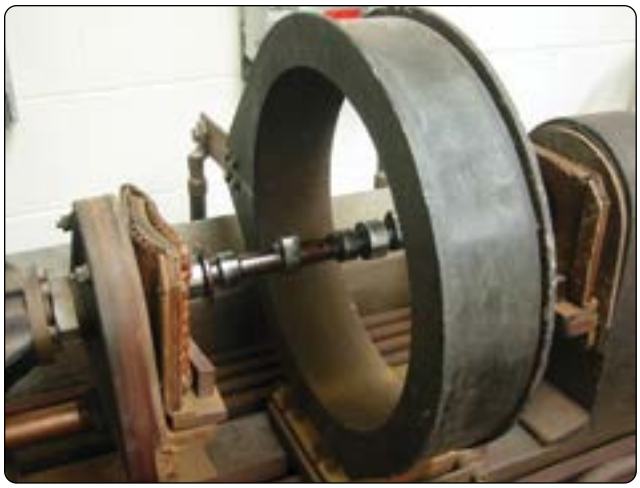

Figure 10-30. *Longitudinal magnetization of camshaft (solenoid method).*

The motor-driven movable head slides horizontally in longitudinal guides and is controlled by a switch. The spring allows sufficient overrun of the motor-driven head to avoid jamming it and also provides pressure on the ends of the work to ensure good electrical contact.

A plunger-operated switch in the fixed head cuts out the forward motion circuit of the movable head motor when the spring has been properly compressed. In some units, the movable head is hand operated, and the contact plate is sometimes arranged for operation by an air ram. Both contact plates are fitted with various fixtures for supporting the work.

The magnetizing circuit is closed by depressing a pushbutton on the front of the unit. It is set to open automatically, usually after about one-half second. The strength of the magnetizing current may be set manually to the desired value by means of the rheostat or increased to the capacity of the unit by the rheostat short circuiting switch. The current utilized is indicated on the ammeter. Longitudinal magnetization is produced by the solenoid that moves in the same guide rail as the movable head and is connected in the electrical circuit by means of a switch.

The suspension liquid is contained in a sump tank and is agitated and circulated by a pump. The suspension is applied to the work through a nozzle. The suspension drains from the work through a nonmetallic grill into a collecting pan that leads back to the sump. The circulating pump is operated by a pushbutton switch.

Portable General Purpose Unit

It is often necessary to perform the magnetic particle inspection at locations where fixed general purpose equipment is not available or to perform an inspection on members of aircraft structures without removing them from the aircraft. It is particularly useful for inspecting landing gear and engine mounts suspected of having developed cracks in service. Portable units supply both AC and DC magnetization.

This unit is a source of magnetizing and demagnetizing current but does not provide a means for supporting the work or applying the suspension. It operates on 200 volt, 60 cycle AC and contains a rectifier for producing DC when required. *[Figure 10-34]*

The magnetizing current is supplied through the flexible cables with prods or contact clamps, as shown in *Figure 10-35*. The cable terminals may be fitted with prods or with contact clamps. Circular magnetization may be developed by using either the prods or clamps.

Longitudinal magnetization is developed by wrapping the

cable around the part. The strength of the magnetizing current is controlled by an eight-point tap switch, and the duration that it is applied is regulated by an automatic cutoff similar to that used in the fixed general purpose unit.

This portable unit also serves as a demagnetizer and supplies high amperage, low-voltage AC for this purpose. For demagnetization, the AC is passed through the part and gradually reduced by means of a current reducer.
In testing large structures with flat surfaces where current must be passed through the part, it is sometimes impossible to use contact clamps. In such cases, contact prods are used.

Prods can be used with the fixed general purpose unit, as well as the portable unit. The part or assembly being tested may be held or secured above the standard unit and the suspension hosed onto the area, while excess suspension drains into the tank. The dry procedure may also be used.

Prods are held firmly against the surface being tested. There is a tendency for a high-amperage current to cause burning at contact areas, but with proper care, such burning is usually slight. For applications where prod magnetization is acceptable, slight burning is normally acceptable.

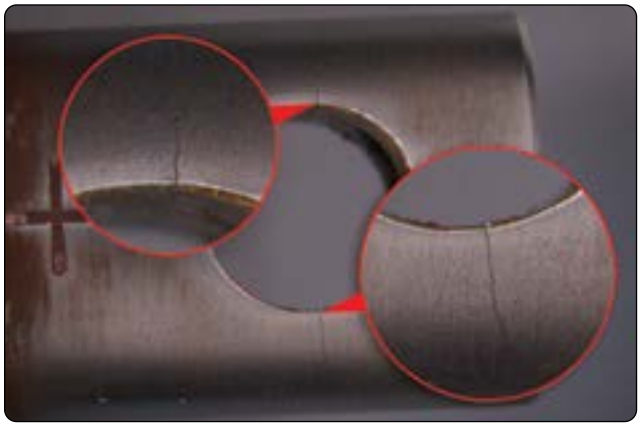

Figure 10-31. *Fatigue crack on the bottom end fitting of a Hydrosorb shock absorber.*

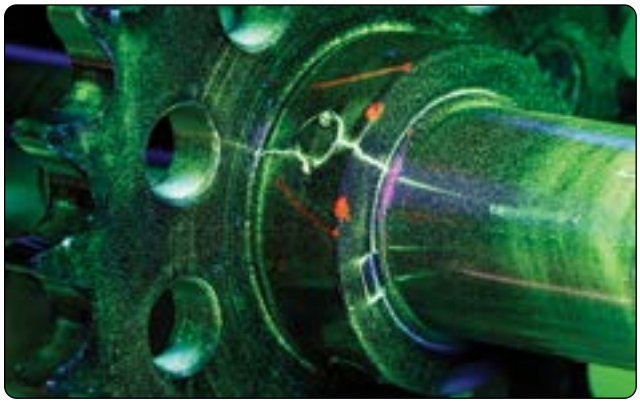

Figure 10-32. *Magnaglo inspection.*

Indicating Mediums

The various types of indicating mediums available for magnetic particle inspection may be divided into two general material types: wet and dry. The basic requirement for any indicating medium is that it produce acceptable indications of discontinuities in parts.

The contrast provided by a particular indicating medium on the background or part surface is particularly important. The colors most extensively used are black and red for the wet procedure and black, red, and gray for the dry procedure.

For acceptable operation, the indicating medium must be of high permeability and low retentivity. High permeability ensures that a minimum of magnetic energy is required to attract the material to flux leakage caused by discontinuities. Low retentivity ensures that the mobility of the magnetic particles is not hindered by the particles themselves becoming magnetized and attracting one another.

Demagnetizing

The permanent magnetism remaining after inspection must be removed by a demagnetization operation if the part is to be returned to service. Parts of operating mechanisms must be demagnetized to prevent magnetized parts from attracting filings, grindings, or chips inadvertently left in the system or steel particles resulting from operational wear. An accumulation of such particles on a magnetized part may cause scoring of bearings or other working parts. Parts of the airframe must be demagnetized so they do not affect instruments.

Demagnetization between successive magnetizing operations

Figure 10-33. *Fixed general-purpose magnetizing unit.*

is not normally required unless experience indicates that omission of this operation results in decreased effectiveness for a particular application. Demagnetization may be accomplished in a number of different ways. A convenient procedure for aircraft parts involves subjecting the part to a magnetizing force that is continually reversing in direction and, at the same time, gradually decreasing in strength. As the decreasing magnetizing force is applied first in one direction and then the other, the magnetization of the part also decreases.

Standard Demagnetizing Practice

The basic procedure for developing a reversing and gradually decreasing magnetizing force in a part involves the use of a solenoid coil energized by AC. As the part is moved away from the alternating field of the solenoid, the magnetism in the part gradually decreases.

A demagnetizer whose size approximates that of the work is used. For maximum effectiveness, small parts are held as close to the inner wall of the coil as possible. Parts that do not readily lose their magnetism are passed slowly in and out of the demagnetizer several times and, at the same time, tumbled or rotated in various directions. Allowing a part to remain in the demagnetizer with the current on accomplishes very little practical demagnetization.

The effective operation in the demagnetizing procedure is that of slowly moving the part out of the coil and away from the magnetizing field strength. As the part is withdrawn, it is kept directly opposite the opening until it is 1 or 2 feet from the demagnetizer. The demagnetizing current is not cut off until the part is 1 or 2 feet from the opening as the part may be remagnetized if current is removed too soon. Another procedure used with portable units is to pass AC through the part being demagnetized, while gradually reducing the current to zero.

Figure 10-34. *Portable magnetic particle inspection equipment.*

Radiographic

Because of their unique ability to penetrate material and disclose discontinuities, X and gamma radiations have been applied to the radiographic (x-ray) inspection of metal fabrications and nonmetallic products.

The penetrating radiation is projected through the part to be inspected and produces an invisible or latent image in the film. When processed, the film becomes a radiograph or shadow picture of the object. This inspection medium and portable unit provides a fast and reliable means for checking the integrity of airframe structures and engines. *[Figure 10-36]*

Radiographic Inspection

Radiographic inspection techniques are used to locate defects or flaws in airframe structures or engines with little or no disassembly. This is in marked contrast to other types of nondestructive testing that usually require removal, disassembly, and stripping of paint from the suspected part before it can be inspected. Due to the radiation risks associated with x-ray, extensive training is required to become a qualified radiographer. Only qualified radiographers are allowed to operate the x-ray units.

Three major steps in the x-ray process discussed in subsequent paragraphs are: exposure to radiation, including preparation; processing of film; and interpretation of the radiograph.

Preparation and Exposure

The factors of radiographic exposure are so interdependent that it is necessary to consider all factors for any particular radiographic exposure. These factors include, but are not limited to, the following:

- Material thickness and density
- Shape and size of the object
- Type of defect to be detected
- Characteristics of x-ray machine used
- The exposure distance
- The exposure angle
- Film characteristics
- Types of intensifying screen, if used

Knowledge of the x-ray unit's capabilities form a background for the other exposure factors. In addition to the unit rating in kilovoltage, the size, portability, ease of manipulation, and exposure particulars of the available equipment must be thoroughly understood. Previous experience on similar objects is also very helpful in the determination of the overall exposure techniques. A log or record of previous exposures provides specific data as a guide for future radiographs. After

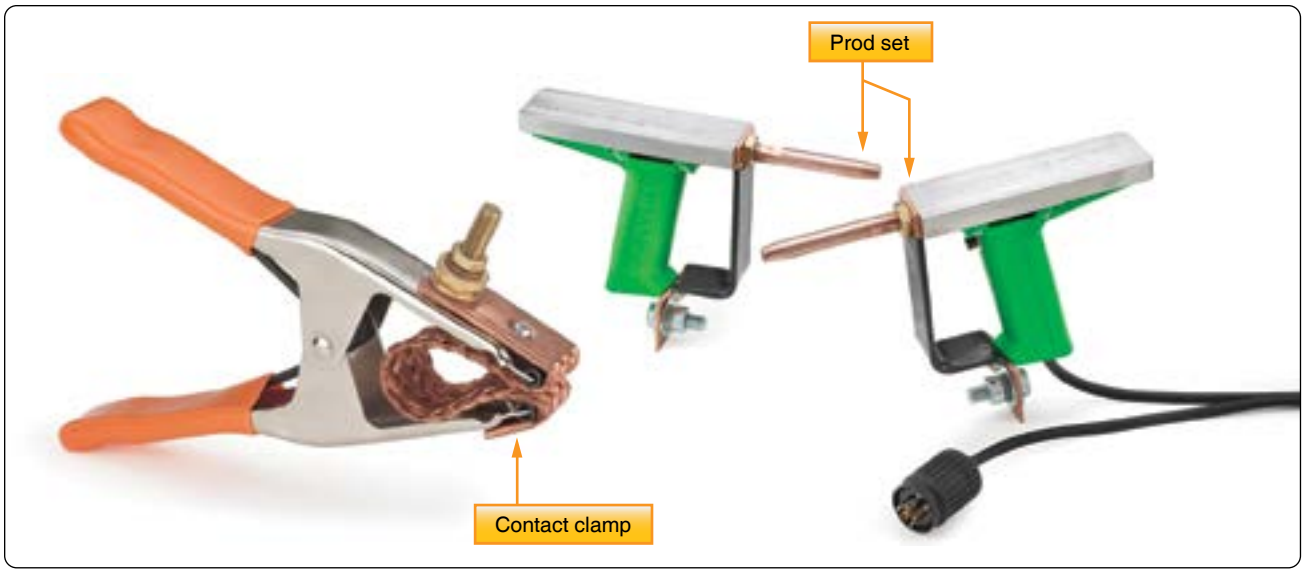

Figure 10-35. *Magnetic particle inspection accessories.*

exposure to x-rays, the latent image on the film is made permanently visible by processing it successively through a developer chemical solution, an acid bath, and a fixing bath, followed by a clear water wash.

Radiographic Interpretation

From the standpoint of quality assurance, radiographic interpretation is the most important phase of radiography. It is during this phase that an error in judgment can produce disastrous consequences. The efforts of the whole radiographic process are centered in this phase, where the part or structure is either accepted or rejected. Conditions of unsoundness or other defects that are overlooked, not understood, or improperly interpreted can destroy the purpose and efforts of radiography and can jeopardize the structural integrity of an entire aircraft. A particular danger is the false sense of security imparted by the acceptance of a part or structure based on improper interpretation.

As a first impression, radiographic interpretation may seem simple, but a closer analysis of the problem soon dispels this impression. The subject of interpretation is so varied and complex that it cannot be covered adequately in this type of document. Instead, this chapter gives only a brief review of basic requirements for radiographic interpretation, including some descriptions of common defects.

Experience has shown that, whenever possible, it is important to conduct radiographic interpretation close to the radiographic operation. When viewing radiographs, it is helpful to have access to the material being tested. The radiograph can thus be compared directly with the material being tested, and indications due to such things as surface condition or thickness variations can be immediately determined. The following paragraphs present several factors that must be considered when analyzing a radiograph.

There are three basic categories of flaws: voids, inclusions, and dimensional irregularities. The last category, dimensional irregularities, is not pertinent to these discussions, because its prime factor is one of degree and radiography is not exact. Voids and inclusions may appear on the radiograph in a variety of forms ranging from a two-dimensional plane to a three-dimensional sphere. A crack, tear, or cold shut most nearly resembles a two-dimensional plane, whereas a cavity looks like a three-dimensional sphere. Other types of flaws, such as shrink, oxide inclusions, porosity, and so forth, fall somewhere between these two extremes of form.

It is important to analyze the geometry of a flaw, especially for items such as the sharpness of terminal points. For example, in a crack-like flaw, the terminal points appear much sharper in a sphere-like flaw, such as a gas cavity. Also, material strength may be adversely affected by flaw shape. A flaw having sharp points could establish a source of localized stress concentration. Spherical flaws affect material strength to a far lesser degree than do sharp-pointed flaws. Specifications and reference standards usually stipulate that sharp-pointed flaws, such as cracks, cold shuts, and so forth, are cause for rejection.

Material strength is also affected by flaw size. A metallic component of a given area is designed to carry a certain load plus a safety factor. Reducing this area by including a large flaw weakens the part and reduces the safety factor. Some flaws are often permitted in components due to these safety factors. In this case, the interpreter must determine the degree

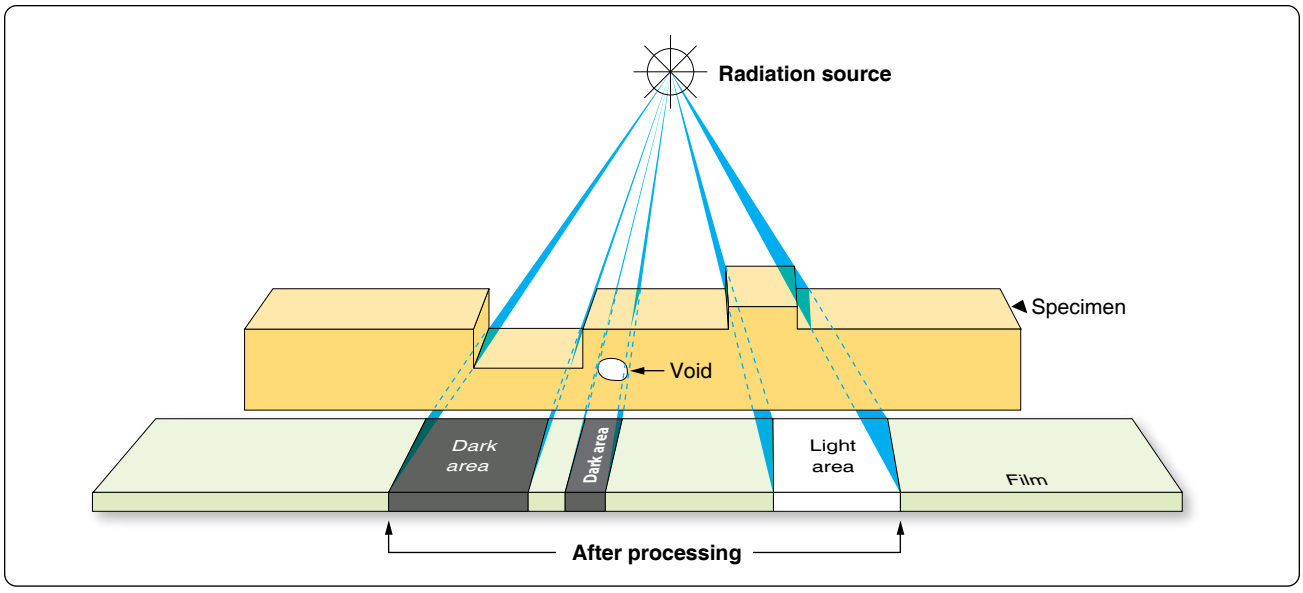

Figure 10-36. *Radiograph.*

of tolerance or imperfection specified by the design engineer. Both flaw size and flaw shape are considered carefully, since small flaws with sharp points can be just as bad as large flaws with no sharp points.

Another important consideration in flaw analysis is flaw location. Metallic components are subjected to numerous and varied forces during their effective service life. Generally, the distribution of these forces is not equal in the component or part, and certain critical areas may be rather highly stressed. The interpreter must pay special attention to these areas. Another aspect of flaw location is that certain types of discontinuities close to one another may potentially serve as a source of stress concentrations creating a situation that must be closely scrutinized.

An inclusion is a type of flaw that contains entrapped material. Such flaws may be of greater or lesser density than the item being radiographed. The foregoing discussions on flaw shape, size, and location apply equally to inclusions and to voids. In addition, a flaw containing foreign material could become a source of corrosion.

Radiation Hazards

Radiation from x-ray units and radioisotope sources is destructive to living tissue. It is universally recognized that in the use of such equipment, adequate protection must be provided. Personnel must keep outside the primary x-ray beam at all times.

Radiation produces change in all matter that it passes through. This is also true of living tissue. When radiation strikes the molecules of the body, the effect may be no more than to dislodge a few electrons, but an excess of these changes could cause irreparable harm. When a complex organism is exposed to radiation, the degree of damage, if any, depends on the body cells that have been changed.

Vital organs in the center of the body that are penetrated by radiation are likely to be harmed the most. The skin usually absorbs most of the radiation and reacts earliest to radiation.

If the whole body is exposed to a very large dose of radiation, death could result. In general, the type and severity of the pathological effects of radiation depend on the amount of radiation received at one time and the percentage of the total body exposed. Smaller doses of radiation could cause blood and intestinal disorders in a short period of time. The more delayed effects are leukemia and other cancers. Skin damage and loss of hair are also possible results of exposure to radiation.

Inspection of Composites

Composite structures are inspected for delamination (separation of the various plies), debonding of the skin from the core, and evidence of moisture and corrosion. Previously discussed methods including ultrasonic, acoustic emission, and radiographic inspections may be used as recommended by the aircraft manufacturer. The simplest method used in testing composite structures is the tap test. Newer methods, such as thermography, have been developed to inspect composite structures.

Tap Testing

Tap testing, also referred to as the ring test or coin test, is widely used as a quick evaluation of any accessible surface

to detect the presence of delamination or debonding. The testing procedure consists of lightly tapping the surface with a light weight hammer (maximum weight of 2 ounces), a coin, or other suitable device. The acoustic response or "ring" is compared to that of a known good area. A "flat" or "dead" response indicates an area of concern. Tap testing is limited to finding defects in relatively thin skins, less than 0.080" thick. On honeycomb structures, both sides need to be tested. Tap testing on one side alone would not detect debonding on the opposite side. *[Figure 10-37]*

Electrical Conductivity

Composite structures are not inherently electrically conductive. Some aircraft, because of their relatively low speed and type of use, are not affected by electrical issues. Manufacturers of other aircraft, such as high-speed, high-performance jets, are required to utilize various methods of incorporating aluminum or copper into their structures to make them conductive. The aluminum or copper (aluminum is used with fiberglass and Kevlar, while copper is used with carbon fiber) is imbedded within the plies of the lay-ups either as a thin wire mesh, screen, foil, or spray. When damaged sections of the structure are repaired, care must be taken to ensure that the conductive path be restored. Not only is it necessary to include the conductive material in the repair, but the continuity of the electrical path from the original conductive material to the replacement conductor and back to the original must be maintained. Electrical conductivity may be checked by use of an ohmmeter. Specific manufacturer's instructions must be carefully followed.

Thermography

Thermography is an NDI technique often used with thin composite structures that use radiant electromagnetic thermal energy to detect flaws. Most common sources of heat are heat lamps or heater blankets. The basic principle of thermal inspection consists of measuring or mapping of surface temperatures when heat flows from, to, or through a test object. All thermographic techniques rely on differentials in thermal conductivity between normal, defect-free areas and those having a defect. Normally, a heat source is used to elevate the temperature of the article being examined while observing the surface heating effects. Because defect-free areas conduct heat more efficiently than areas with defects, the amount of heat that is either absorbed or reflected indicates the quality of the bond. The type of defects that affect the thermal properties include disbonds, cracks, impact damage, panel thinning, and water ingress into composite materials and honeycomb core. Thermal methods are most effective for thin laminates or for defects near the surface.

The most widely used thermographic inspection technique uses an infrared (IR) sensing system to measure temperature distribution. This type of inspection can provide rapid, one-sided, non-contact scanning of surfaces, components, or assemblies. The heat source can be as simple as a heat lamp, so long as the appropriate heat energy is applied to the inspection surface. The induced temperature rise is a few degrees and dissipates quickly after the heat input is removed. The IR camera records the IR patterns. The resulting temperature data is processed to provide more quantitative information. An operator analyzes the screen and determines whether a defect was found. Because IR thermography is a radiometric measurement, it can be done without physical contact. Depending on the spatial resolution of the IR camera and the size of the expected damage, each image can be of a relatively large area. Furthermore, as composite materials do not radiate heat nearly as much as aluminum and have higher emissivity, thermography can provide better definition of damage with smaller heat inputs. Understanding of structural arrangement is imperative to ensure that substructure is not

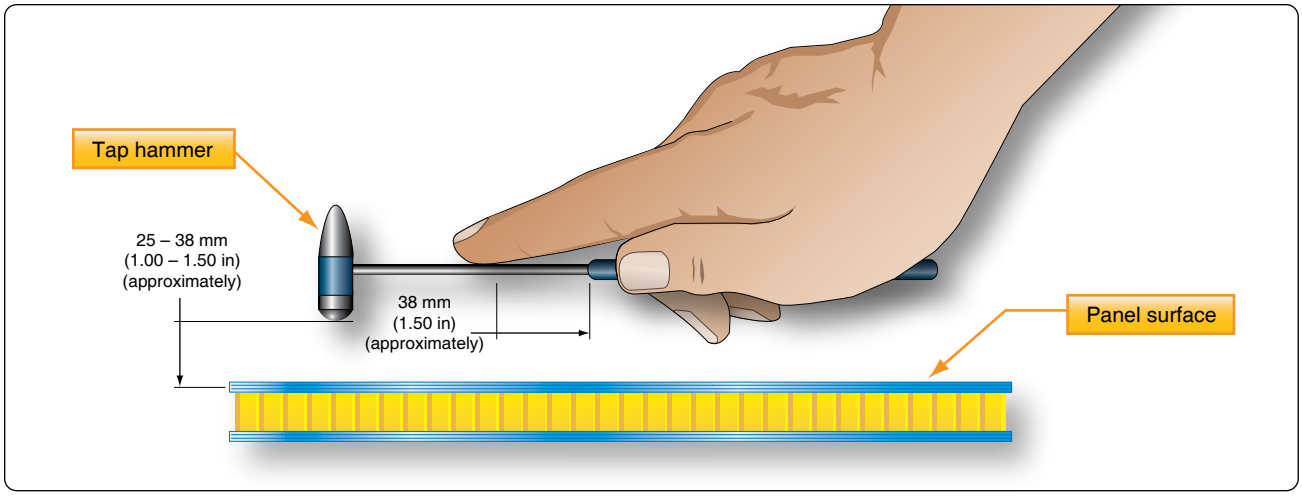

Figure 10-37. *Tap testing using hammer.*

mistaken for defects or damage.

Inspection of Welds

A discussion of welds in this chapter is confined to judging the quality of completed welds by visual means. Although the appearance of the completed weld is not a positive indication of quality, it provides a good clue about the care used in making it. A properly designed joint weld is stronger than the base metal that it joins. The characteristics of a properly welded joint are discussed in the following paragraphs.

A good weld is uniform in width; the ripples are even and well feathered into the base metal and show no burn due to overheating. *[Figure 10-38]* The weld has good penetration and is free of gas pockets, porosity, or inclusions. The edges of the bead are not in a straight line, yet the weld is good since penetration is excellent.

Penetration is the depth of fusion in a weld. Thorough fusion is the most important characteristic contributing to a sound weld. Penetration is affected by the thickness of the material to be joined, the size of the filler rod, and how it is added. In a butt weld, the penetration should be 100 percent of the thickness of the base metal. On a fillet weld, the penetration requirements are 25 to 50 percent of the thickness of the base metal. The width and depth of bead for a butt weld and fillet weld are shown in *Figure 10-39*.

To assist further in determining the quality of a welded joint, several examples of incorrect welds are discussed in the following paragraphs.

The weld in *Figure 10-38A* was made too rapidly. The long and pointed appearance of the ripples was caused by an excessive amount of heat or an oxidizing flame. If the weld were cross-sectioned, it would probably disclose gas pockets, porosity, and slag inclusions.

Figure 10-38B illustrates a weld that has improper penetration and cold laps caused by insufficient heat. It appears rough and irregular, and its edges are not feathered into the base metal. The puddle tends to boil during the welding operation if an excessive amount of acetylene is used. This often leaves slight bumps along the center and craters at the finish of the weld. Cross-checks are apparent if the body of the weld is sound. If the weld were cross-sectioned, pockets and porosity are visible. *[Figure 10-38C]*

A bad weld has irregular edges and considerable variation in the depth of penetration. It often has the appearance of a cold weld.

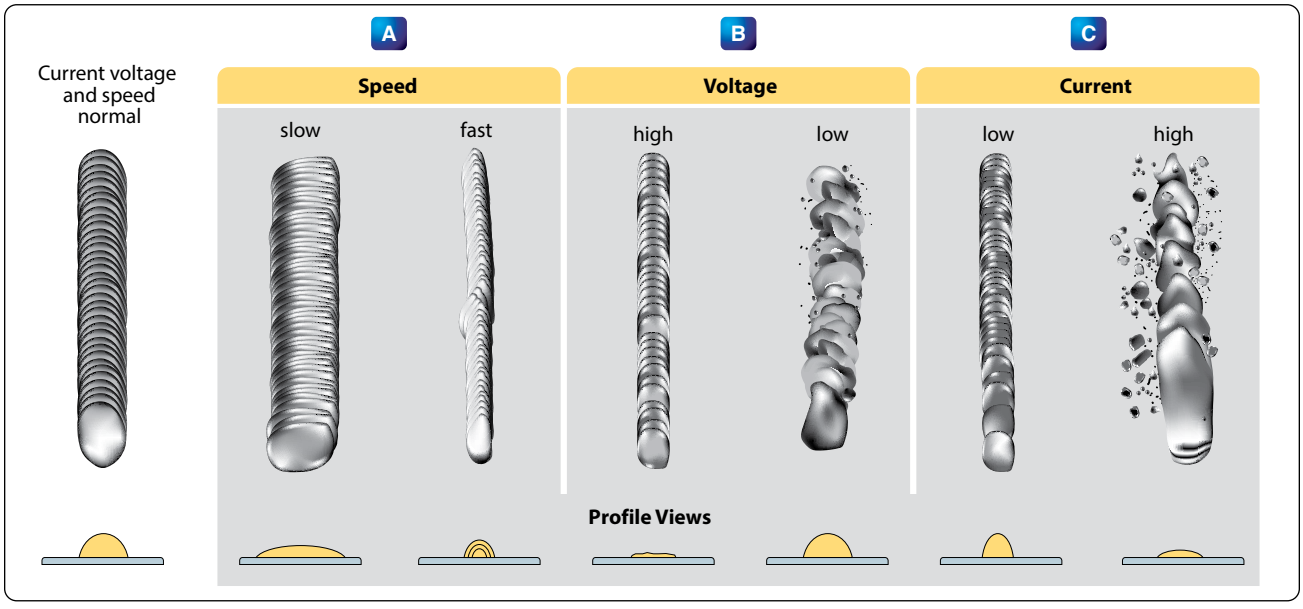

Figure 10-38. *Examples of poor welds: too rapidly (A), improper penetration and cold laps (B), and irregular edges and considerable variation (C).*

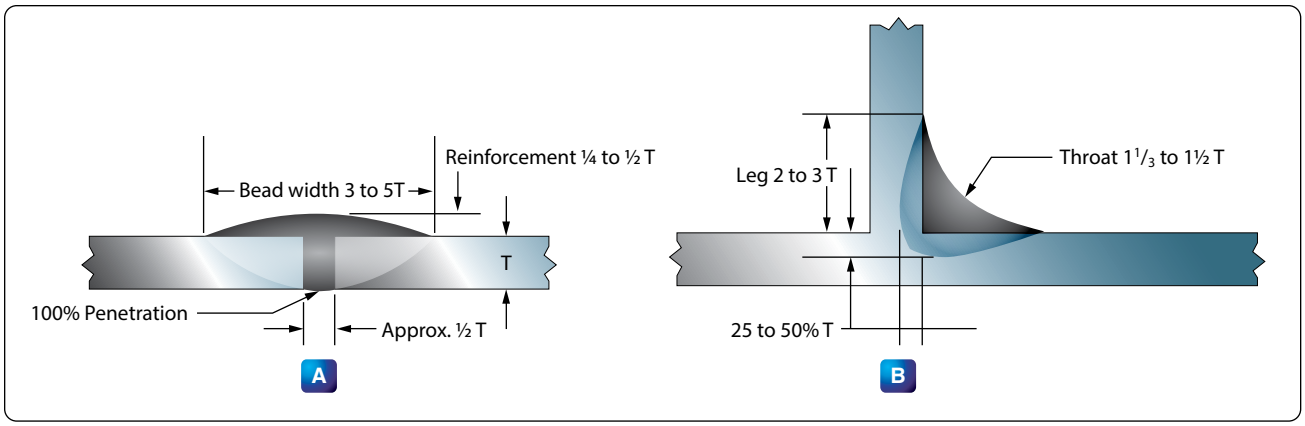

Figure 10-39. *Butt weld (A) and fillet weld (B), showing width and depth of bead.*